Supplement to: Channel Estimation for Rydberg Arrays With Learned Filtering and Recovery Checks

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***Abstract*—This supplement specifies the final receiver's policy, algebraic validity conditions, development/confirmation protocol, adverse cases and numerical evidence. It then preserves the classical and learned studies from which the receiver developed. Historical experiments retain their own models, data, metrics and inference procedures; statements about a historical primary method or final bank refer only to that study. The new receiver is the matched-context hybrid with the confirmed residual-triggered algebraic fallback. No historical table is pooled with its fresh 153-condition comparison. The supplementary evidence is a summary and provenance archive, not a full public raw-data and checkpoint release.**

I. Final Receiver: Stored Components and Exact Policy

The current receiver uses the matched-30-context adaptive and global fits for seeds 9101, 9102 and 9103. One paired inference fit stores 218 neural parameters; reported fit means average metrics and do not ensemble channels. Its observation inputs are Z, S, B, v, with $N = 32$, $K = 3$, $P = 30$ in the final experiment. The complete policy and checkpoint/source hashes are supplied in `evidence/reliability/source/policy.json` and the protocol record. True channel, path count and test error are not routing inputs.

Let $s = 10\log_{10}(K/v)$ and $r = 10\log_{10}\operatorname{mean}|B|^2$. Reference keys are nearest to 5, 12 and 20 dB; SNR keys are nearest to the declared 17-point grid. Exact ties select the lower key. A tolerance of 10^{-10} dB is used at the continuous boundaries. The adaptive start is used for $s \le 30 + 10^{-10}$, the global schedule above it. At $s \le 20 + 10^{-10}$, return the raw adaptive output exactly. Elsewhere use Table I. If the family's lookup record disagrees with continuous model routing, extra EM is set to zero. It is a deterministic off-grid rule, not interpolation or a guarantee of smooth performance.

The final filter applies the same smooth Hankel map as the trained stages with $\beta = 1$ and

$$\tau_k = c\sqrt{2v\,\mathrm{Re}[(SS^H)^{-1}]_{kk}\max(m,q)}. \tag{1}$$

For $c = 0$, it returns its input exactly. Positive c gives a decreasing threshold as supplied noise decreases; it is a heuristic scale, not a theorem about the noise covariance after EM. Maintaining this filter above 40 dB removed an earlier artificial switch-off, while other piecewise transitions remain.

Only cases in the nearest-reference 5 dB bin with $s > 30 + 10^{-10}$ enter the new diagnostic. The complete incumbent G_0 is computed first. The maximum antenna residual

$$\max_n \frac{2}{Pv}\sum_p [Z_{np} - |[G_0 S + B]_{np}|]^2 \tag{2}$$

exceeding 2 triggers one algebraic start, 20 exact EM updates and the same final-filter rule. The complete fallback replaces the incumbent only if finite and its average fitting Rician negative log-likelihood is smaller by more than 10^{-12}. Failed solves, no triggers and rejected alternatives return the incumbent bitwise. The standalone algebraic-start-plus-EM20 baseline instead applies the algebraic start on every condition and uses spectral-plus-EM20 when that solve is invalid.

II. Intensity Regression: Coefficients, Noise and Scope

For repeated $b_n \neq 0$, define $c_n = g_n/b_n$ and $y_{np} = (Z_{np}^2 - v)/|b_n|^2 - 1$. The expansion is

$$\begin{aligned}\mathbb{E}[y_{np}] = {} & 2\,\mathrm{Re}\sum_i c_{ni}s_{ip} + \sum_i |c_{ni}|^2|s_{ip}|^2 \\ & + 2\,\mathrm{Re}\sum_{i<j} c_{ni}c_{nj}^* s_{ip}s_{jp}^*.\end{aligned} \tag{3}$$

All expectations here condition on G, S, B. The corresponding real coefficients are the real/imaginary components of c_n, its three squared magnitudes and the real/imaginary components of each pair product. The signs of the imaginary columns are negative because $\mathrm{Re}(ab) = \mathrm{Re}\,a\,\mathrm{Re}\,b - \mathrm{Im}\,a\,\mathrm{Im}\,b$. There are $d = 2K + K^2 = 15$ columns for three users. The real least-squares solution treats the quadratic coefficients as nuisance variables; it does not enforce their consistency with a rank-one outer product.

Writing $y_n = D\xi_n + \epsilon_n$ gives $\mathbb{E}\epsilon_n = 0$ and

$$\mathrm{var}(\epsilon_{np}) = \frac{2v|g_n^T s_p + b_n|^2 + v^2}{|b_n|^4}. \tag{4}$$

Independent field noise makes this conditional covariance diagonal over pilots. If D has full column rank, $\widehat{\xi}_n = (D^TD)^{-1}D^T y_n$ is conditionally unbiased and has covariance

$$(D^TD)^{-1}D^T\,\mathrm{cov}(\epsilon_n)D(D^TD)^{-1}. \tag{5}$$

These formulas explain sensitivity to conditioning and noise/reference scale. The ordinary least-squares solve is not asserted to be minimum variance. At $v = 0$, $\epsilon_n = 0$ and full rank recovers c_n exactly. In floating-point tests, the independent noiseless algebraic reconstruction errors were of order 10^{-15}.

The implementation uses SVD least squares, checks rank 15 and condition number at most 10^8, and rejects nonfinite coefficients or a reference that does not repeat exactly. With constant-modulus pilots, each diagonal design column is a constant vector, so three such columns span at most one dimension and rank is at most 13. The current validity rule

TABLE I
FROZEN PRIOR-HYBRID TABLE INHERITED BY THE FINAL RECEIVER. ENTRIES ARE EXTRA EM UPDATES AND FINAL NOISE-SCALE MULTIPLIER c. REF DENOTES NEAREST-REFERENCE KEY. EM STARTS FROM THE CHOSEN TRAINED OUTPUT. AT SNR AT MOST 20 DB BOTH CONTINUATION AND FINAL FILTERING ARE BYPASSED.

SNR key (dB)	T, ref 5	T, ref 12	T, ref 20	c, ref 5	c, ref 12	c, ref 20
0–30	0	0	0	0	0	0
32.5	0	10	10	0	2	2
35	0	10	40	0	2	4
37.5	0	20	40	0	4	4
40, 42.5, 45	40	20	40	0	4	4
50, 60	40	40	40	0	4	4

then rejects. A reduced parameterization could remove this redundancy, but is not implemented or tested here. Extra offline known QPSK probes and online QPSK data must not be confused with the Gaussian fitting pilots. Neither unbiasedness of the unselected initializer nor noiseless exactness extends automatically to the refined, filtered, selected receiver.

III. TRAINING, OFFLINE SELECTION AND INDEPENDENT CONFIRMATION

The final adaptive and global models train on the same 39-observation blocks, using 30 fitting observations and 9 held out. Training/validation block counts are 16,384/2,048, batch size 32, epochs 20 and Adam learning rate 0.001. The validation score equally averages 126 SNR/reference/path strata. All three fit seeds share this data population; model fits and data-population replications are different uncertainties. The older 21-context and re-corruption experiments retained later in this supplement are separate recipes.

Each new policy bank A/C has 768 independent physical anchors (256 in each of three path regimes), reference 5 dB and eight coupled SNRs above 30 dB. It supplies 30 context observations, 9 independent held-out Gaussian observations and 64 known-symbol probes per channel. Ground-truth channels are used by the simulator but are not exported to development selection. The 12 rules cross two residual summaries, thresholds 2/4/8 and tails 10/20. All pass the separate complete-pipeline calibration cap of 10 ms p90; 144 forced-fallback calls separately exercise longer branches.

Selection requires positive held-out-loss gain and empirical probe-error nonincrease in every path regime. The incumbent is always admissible, and tie rules are fixed. The chosen maximum/2/20 rule triggers on 27.848% of A cases and 27.837% of C cases, averaging fits. C's one-sided 95% lower loss-gain bound is 0.198729; its descriptive two-sided 95% interval is [0.195729, 0.370585]. Both methods have zero observed probe errors in each regime. Only the selected rule is confirmed; no substitute is searched on C or the final test. This confirms the incremental wrapper under the protocol, not every earlier inherited EM-tail decision.

The current fresh test uses 1,152 independent anchors in nine reference/path blocks and 19,584 coupled SNR condition cases. Data BER uses 7,520,256 bits per fit. The bootstrap shares within-block indices over SNR, methods and fits and draws independently across blocks. The four whole-grid outcome intervals are multiplicity adjusted; regional and per-cell intervals remain descriptive. The evidence archive preserves all raw counts and per-fit summaries available in the compact source records.

IV. EXACT BASELINE DEFINITIONS AND FROZEN SELECTION

The directory `evidence/reliability/baselines/` contains byte-for-byte copies of the inherited baseline policies and their selection records. `PROVENANCE.json` gives each original repository path, SHA-256 digest and copy size, plus the implementation-source hashes. `policy_summary.json` expands all 51 reference/SNR entries into the compact configuration and the full-repair base/filter combination. This is a reporting transformation of frozen records, not a new selection. Baseline parameters were not reoptimized on the final reliability development or test banks.

A. Classical Structured Start and Common-40 Controls

The classical start uses the same spectral initializer as the learned starts, followed by ten exact EM updates with unit relaxation. Each update is followed by smooth Hankel filtering and mixing. Its single global configuration was selected from 30 choices: relative thresholds $\{0.01, 0.03, 0.1, 0.3, 1\}$ or noise multipliers $\{0.5, 1, 2, 4, 8\}$, each crossed with mixing $\{0.1, 0.5, 1\}$. Relative thresholds multiply the current Hankel RMS; noise thresholds use the formula in the final-filter section. The criterion is the minimum raw-start held-out Rician loss, averaged equally over the 126 strata of the same 2,048 validation blocks and fixed 30/9 split used for the learned checkpoint selection. No extra EM continuation or post-start final filter enters this selection score. Evaluation uses double precision on the stored single-precision observations.

The selected start has noise multiplier $c = 1$ and mixing $\beta = 1$ at every stage. The complete classical-start-plus-EM40 control then adds 40 unfiltered exact EM updates and the final-filter table in Table I. The adaptive-start-plus-EM40 and global-schedule-plus-EM40 controls use the corresponding trained start with these same 40 updates and final filter, including at ordinary SNR; they do not use the hybrid's ordinary-SNR bypass. `classical_selection.json` retains all 30 candidate scores and their 126 stratum scores. The copied start-ablation protocol records this selection; the start-budget protocol/policy records the subsequent common-40

comparison. Selection on raw-start likelihood does not guarantee an optimal configuration for the final refined pipeline or for NMSE.

B. Compact Sparse/EM Control

Let A_D have entries $[A_D]_{nj} = N^{-1/2}\exp[-in(-\pi + 2\pi j/D)]$, with $n = 0,\dots,N-1$ and $j = 0,\dots,D-1$, and write $G = A_D X$. The penalized control uses a finite-iteration solver for

$$\frac{v}{2}\sum_{n,p}\ell(Z_{np},[A_D XS + B]_{np},v) + \lambda\sum_{j,k}|X_{jk}|, \qquad \lambda = f\sqrt{vP\log D}. \tag{6}$$

It initializes G spectrally and $X = (N/D)A_D^H G$, computes the exact Rician posterior field at each outer step, and performs the declared number of monotone-restarted FISTA inner updates on the penalized quadratic surrogate. The step size is $[(D/N)\lambda_{\max}(SS^H)]^{-1}$. Coefficients carry between outer steps; finite inner iteration counts are not exact penalized M-step solves. FFT products implement this uniformly spaced spatial-frequency dictionary. The zero-penalty option directly performs 100 unregularized EM updates and does not run a redundant sparse solve.

The compact menu contains three $(D, \text{inner}, \text{outer})$ structures, $(128, 8, 12)$, $(256, 8, 12)$ and $(256, 2, 40)$, each with $f \in \{0.03, 0.1, 0.3, 1, 3\}$, plus direct EM100. At the last sparse refresh, all 16 passed the historical complete-pipeline calibration cap of 10.7439662 ms, defined by the contemporaneous full-repair pooled p90. Only the ten bins at references 12/20 dB and SNR 40/42.5/45/50/60 dB were refreshed: 384 development anchors per path regime supplied 30 fitting and nine independently drawn held-out pilots. Minimum equally weighted held-out Rician loss selected the configuration; exact score ties used calibration p90 then candidate name. The other 41 bins retained the preceding frozen policy. These historical eligibility times are not substituted for the final experiment's fresh latency measurements, and this compact menu is not a globally optimized classical accuracy–time frontier.

`filter_continuation_policy.json` includes every compact configuration and the last refresh's candidate scores; its copied protocol and calibration preserve the menu, bank hashes and timing evidence. The final table has 34 penalized entries and 17 direct-EM100 entries. Nearest-reference/SNR lookup chooses the lower bin at a tie; there is no online rank search or validation split. All 30 deployment pilots are used to estimate the channel.

C. Earlier Full Repaired Receiver

The full-repair comparator uses the matched-context adaptive start, not the global schedule. At reference 5 dB it returns that start through SNR 32.5 dB, adds five exact EM updates at 35 dB, and adds 100 from 37.5 dB upward. At reference 12 dB it returns the adaptive start through 30 dB; at 32.5/35 dB it adds 2/5 EM updates and final filtering with $c = 2$; from 37.5 dB upward it uses spectral EM100 followed by $c = 4$ filtering. At reference 20 dB it returns the adaptive start through 30 dB, adds 3 EM updates and $c = 2$ filtering at 32.5 dB, and uses spectral EM100 with $c = 4$ from 35 dB upward. Final mixing is one, and all thresholds use the supplied variance. These are grid-key descriptions; off-grid inputs use the same nearest-bin rule. The copied filter-continuation policy stores the outer repair choices, while `dual_guard_policy.json` identifies the base beneath each final shrink operation. The native-runtime policy/protocol verifies that numerical optimization preserved these choices and the compact table. Neither baseline acquires the final receiver's algebraic recovery check.

V. Final Effects, Adverse Conditions and Timing

The final primary and regional tables are in the main manuscript. Here the extended effects retain fit dependence and adverse comparisons. No condition-mean NMSE or BER worsens against the prior hybrid in this bank. Against the full repair, 32 NMSE means are worse, with 8 pointwise intervals excluding zero; these are not simultaneous discoveries after correction over 153 cells. The worst loss, 0.700957 dB, is at two paths, reference 20 dB, SNR 32.5 dB, outside the new gate. Compact sparse/EM remains better for aggregate BER and pooled linear NMSE. Zero target-region bit errors do not prove equivalence or exclude rare failures.

Each timing window contains 27 independent physical anchors reused across SNR, giving 459 condition cases and 8,721 calls. Every raw/family quantile, ordering and pass flag was independently checked. The second window is slower for the baselines as well as the receiver. All three receiver fits pass the full-grid and target p90 requirements in both windows, and none of the 51 reference/SNR-bin p90 values exceeds 10.74 ms. Individual timing is different: one of 1,377 receiver calls in window 2 took 22.275833 ms. It was fit 9103, random 3–7 paths, reference 12 dB, SNR 45 dB, anchor index 600, with its gate false. The measurement does not establish a cause. No timing window was rerun to pass.

VI. Development Sequence and Interpretation

The following sequence records what motivated the final design. Scores from its different banks are not a common leaderboard. Detailed report paths and hashes are in `historical_report_index.json` inside the evidence folder.

1) *Runtime-matched sparse comparison.* An angle-domain Rician-likelihood baseline was adapted and selected under a processing budget, strengthening the classical comparator. Its final frozen form includes a direct-EM option.
2) *Pilot-matched retraining and short EM.* Comparing 30 versus 21 context pilots on paired 39-observation blocks gives 0.219 dB NMSE gain and 0.01758 percentage-point BER reduction on that separate test, with adjusted intervals [0.205, 0.233] dB and [0.01303, 0.02245] percentage points. All selected ordinary-SNR extra tails are zero; fixed 1/2/5-step continuations harm the tested adaptive estimate. The study supports matching training inputs, not a generic claim that more EM helps.

TABLE II
PAIRED WHOLE-GRID IMPROVEMENTS RELATIVE TO EACH REFERENCE. POSITIVE VALUES FAVOR THE RELIABILITY RECEIVER.

Reference	Interval	NMSE gain (dB)	Interval (dB)	BER reduction (pp)	Interval (pp)
Prior selective EM	Adjusted	0.507	[0.338, 0.652]	0.000000	[0.000000, 0.000000]
Full repair	Adjusted	0.762	[0.231, 0.878]	0.000075	[0.000000, 0.000266]
Compact sparse/EM	Pointwise	6.725	[5.982, 7.103]	-0.033337	[-0.038164, -0.028612]
Adaptive + EM40	Pointwise	2.929	[2.530, 2.994]	0.285708	[0.276258, 0.295150]
Global + EM40	Pointwise	2.138	[2.076, 2.203]	0.291222	[0.281662, 0.300876]
Classical + EM40	Pointwise	6.615	[5.659, 7.023]	0.286892	[0.277353, 0.296422]
Spectral EM100	Pointwise	8.058	[6.900, 8.500]	0.306333	[0.296501, 0.316262]
Algebraic + EM20	Pointwise	2.087	[2.002, 2.179]	0.313181	[0.303354, 0.323074]

Adjusted rows use 98.75% percentile intervals (Bonferroni across two contrasts and two outcomes for nominal 95% family coverage). Other rows use descriptive 95% intervals. The 20,000 resamples are paired within nine independent reference/path blocks; channel indices are shared across SNRs and fits within a block. Inference is conditional on the fixed fits, policy, and simulation grid. A collapsed empirical BER interval does not establish equivalence or noninferiority.

TABLE III
CONDITION-LEVEL ADVERSE RESULTS OUT OF 153 CONDITIONS.

Reference	NMSE worse	NMSE harm CI	BER worse	BER harm CI	Worst NMSE gain (dB)
Prior selective EM	0	0	0	0	0.000
Full repair	32	8	0	0	-0.701
Compact sparse/EM	22	20	22	12	-1.214
Adaptive + EM40	29	12	3	0	-3.747
Global + EM40	20	10	3	0	-3.736
Classical + EM40	24	2	3	0	-0.700
Spectral EM100	3	2	3	0	-0.701
Algebraic + EM20	44	15	3	0	-3.746

Worse means a negative point improvement; harm CI means a descriptive pointwise 95% interval lies below zero. Complete adverse condition identifiers, fit effects, and intervals are retained in the archived summaries. No multiplicity-adjusted subgroup safety claim is made.

TABLE IV
INDIVIDUAL RECEIVER TIMING EXTREMES, RETAINING BOTH WINDOWS.

Window	Maximum (ms)	> 10.74 ms	Fit	Paths	RSR/SNR (dB)	Index	Gate
Window 1	5.225209	0/1377	9102	Random 3–7	5/50	500	false
Window 2	22.275833	1/1377	9103	Random 3–7	12/45	600	false

The second window contains one 22.275833 ms call despite satisfying both p90 targets. Its gate was false; these measurements do not establish the cause. Pooled p90 eligibility is not a hard per-call deadline.

3) *Retrained high-SNR bypass.* A broad high-SNR gain still leaves eight adverse conditions, with worst loss 16.415 dB. A single coarse bypass is insufficient near weak-reference transitions.
4) *Weak-reference transition and incumbent-preserving repairs.* Subsequent transition, incumbent and dual-guard studies reduce specific failures but reveal regressions against earlier versions. They are development evidence for candidate generation and acceptance logic, not proof of a uniformly better filter. The first transition repair improves its weak 18-condition region by 16.208 dB but still has 10 whole-grid NMSE losses and misses its target-region p90 cap at 11.258 ms.
5) *Filter continuation above 40 dB.* Keeping the strong-reference noise-scaled filter active changes 30 strong/high conditions while preserving 123 others. Its separate mean shifts from -53.364 to -58.557 dB with zero observed target BER in both. This diagnoses an artificial filter-removal jump.
6) *Native runtime repair.* Numerical-kernel optimization preserves the same formulas, with recorded coefficient discrepancy at most 6.66×10^{-14} in its parity checks. It is an implementation improvement, not a new estimator-accuracy result.
7) *Matched starts and common continuation.* Adaptive, global and classical structured starts were compared with common EM/filtering. In the common 40-tail comparison, adaptive starts lose 0.696 dB to the global schedule on mean NMSE, while improving BER by 0.005984 percentage points. Forced 40-step refinement loses 4.808 dB in ordinary-SNR adaptive performance. This motivates preserving the ordinary branch and using different high-SNR effort.
8) *Hybrid routing and selective tails.* The zero-tail hybrid underperforms the full repair. It was retained after independent confirmation rejected all proposed nonzero

tails. Exploratory nonzero-tail proposals improve that hybrid, but all 21 earlier C confirmation decisions reject the proposed changes. The final reliability experiment explicitly adopts that frozen exploratory prior as its incumbent; its new confirmation does not erase these failures.

9) *Current reliability wrapper.* A new, independently calibrated residual gate and different initializer are confirmed before the fresh final test. Only this current experiment supplies the main manuscript's final nine-method comparison and timing verdict.

VII. Evidence Integrity and Reproduction Scope

The source bundle includes compact final protocols, policy, source/checkpoint hashes, all-method summaries, adverse-cell lists, timing records, five audits, figure values and 163 independently checked numerical table entries. It excludes the large raw NPZ banks and neural checkpoint binaries. Compilation requires only the included figures, bibliography and TeX sources. Regenerating simulations requires the original code/data environment or a future complete public release; no archive DOI has been invented.

The final selection audit reconstructed development observations and decisions before test generation. Independent base and fallback prediction replays match exactly. The test audit replays 2,448 standalone predictions, independently rebuilds 9,216 gate/fit cases and verifies ordinary output preservation. The BER audit independently checks 6,120 decodes, 306 data reconstructions and 915 exact prediction/count reuses. The summary audit checks the nine-block dependence, all 18 scopes and adjusted intervals; independent bootstrap arithmetic differs by at most 2.13×10^{-14}. The timing audit checks 17,442 complete calls. A development-probe reconstruction check uses explicit 10^{-13} tolerances for independent floating-point arithmetic, with maximum difference 8.88×10^{-16}; exact production-preservation checks remain exact.

Frozen study files and checkpoints were not edited for this manuscript. Manuscript figures and tables are reporting-only transformations of verified evidence. Current final policies were not chosen from fresh test errors. Earlier development choices and known limitations are retained rather than relabelled as independent confirmation.

VIII. Additional Implementation Details

These details support the shorter main paper. They preserve the initialization, older baseline searches and numerical settings. They do not introduce new experiments.

A. Shared Spectral Starting Estimate

The learned and direct-EM starts use the following spectral initialization. The algebraic recovery start is different. For row n, form C_n with rows $[s_p^T, b_{np}]$ and take the leading eigenvector u_n of $C_n^H \operatorname{diag}(z_n) C_n$. Set

$$c_n = \frac{\sum_p z_{np} |(C_n u_n)_p|}{\sum_p |(C_n u_n)_p|^2}. \tag{7}$$

The first K entries of $c_n u_n e^{-j \arg(u_{n,K+1})}$ initialize the channel row. When observations are withheld, only the fitting subset enters initialization. Otherwise the estimator would contain information from its validation observations.

B. Earlier OptShrink Comparisons

OptShrink instead estimates retained weights from the discarded spectrum [1]. For singular values $s_1 \geq \cdots \geq s_q$ and candidate rank $r < q$, define

$$\phi_d(z) = \frac{1}{d-r}\left[\sum_{j=r+1}^{q} \frac{z}{z^2 - s_j^2} + \frac{d-q}{z}\right],$$
$$\mathcal{D}(z) = \phi_m(z)\phi_q(z), \quad \widehat{s}_i = -2\mathcal{D}(s_i)/\mathcal{D}'(s_i), \ i \leq r. \tag{8}$$

Other weights are zero. We rescale the spectrum, guard unresolved poles, average antidiagonals and mix the result with the measurement update. The adaptation is applied either after every update or once at the end.

In the historical OptShrink study, the broad offline search uses the original learned estimator's 2,048 validation observations and the same 21/9 masks. It covers both placements, ranks 1–15 plus no filtering, mixing factors $0.25, 0.5, 0.75, 1$, and update budgets $5, 10, 20, 50, 100$: 610 combinations. Rician validation loss is averaged equally over six path regimes. Settings are frozen for each of 21 noise/reference pairs before testing. Path labels balance offline selection; they are not inference inputs. Off-grid stress tests use the nearest lookup without retuning.

A stronger in-loop control selects rank for each received block. It keeps the lookup's mixing and iteration budget, fits ranks 1–15 plus bypass to 21 pilots, scores nine held-out pilots with Rician loss, and restarts on all 30. All search and refitting time counts. This is not an unrestricted search over user-specific ranks, mixing factors and stopping times. It uses no extra pilot observations, but pays computation to use the current block rather than only offline averages.

C. Numerical Rules

For reproducibility, channel power and singular-value sum are floored at 10^{-12}, and squared Hankel RMS scale at 10^{-24}. Pilot-Gram systems use direct solves without an added ridge. PyTorch implementations use scaled Bessel functions and a series for $\kappa^2 < 10^{-4}$. OptShrink zeros weights at relative squared spectral gaps at most 10^{-13}. These are fixed numerical rules, not tuned extra physical assumptions. Positive threshold makes the reconstruction solve smooth, although ordered spectral features need not be differentiable at every tie.

D. Earlier Classical Test Settings

We keep distinct studies separate. The classical array study uses $K = 3$, $N = 8, 16, 32$, $P = 10, 30$, RSR 12 dB, six SNRs from -5 to 20 dB and 400 paired worlds per cell. Unless fixed, user path counts are independently uniform on 3–7. It rejects circular spatial-frequency separation below 10^{-10}, steering-matrix singular-value ratio below 10^{-8} and pilot singular-value

ratio below 10^{-8}. Effective-rank studies use $N = 16, 32, 64$, $P = 20$, RSR 10 dB and 300–400 worlds per cell with SNR uniform on $[-10, 20]$ dB. The smooth/hard-filter comparison uses $N = 32$, $P = 30$, RSR 12 dB, $L = 3, 7, 14$ and SNR 0 and 10 dB, with 400 worlds per cell. These screened classical samples and their information references are not pooled with the final receiver results.

E. Complete Timing Procedure and Operation Counts

Two isolated timing windows measure complete batch-one inference on one CPU thread, in double precision, with randomized interleaving of all 19 individual methods/fits. Each window has 27 independent physical anchors (three per reference/path block), reused across SNR to form 459 condition cases and 8,721 calls. The first follows policy freeze and precedes fresh testing; the second follows all accuracy jobs. Initialization, conversions, features, solves, continuation, final filtering, residual checks and accepted or rejected fallback work are included. We exclude offline training/tuning, model loading, preloaded shape constants, observation generation and BER decoding. The optimized numerical kernels use NumPy/SciPy with PyTorch interfaces; historical all-PyTorch times are not substituted. Both whole-grid and target-region p90 must be at most the predeclared 10.74 ms software-processing target in both windows. This is a host-specific percentile target, not a per-symbol hardware deadline. Neither window is rerun to obtain a pass.

An adaptive stage costs $O(NKP + NK^2 + K^3 + Kmq^2 + Kq^3)$ in the dense implementation. The global schedule avoids adaptive feature/controller evaluation but still uses measurement updates and structural solves. With $d = K^2 + 2K$, the algebraic branch requires a real design solve of order $O(Pd^2 + d^3 + NPd)$, followed by its EM continuation; here $d = 15$. Neural parameter count alone does not capture either online latency or offline data/calibration cost. The complete paired fit stores 218 neural parameters, plus policy entries, and uses extra held-out observations and known symbols during offline policy development.

IX. Bit Error Rate Over the Full SNR Range

The main paper shows 0–30 dB for readability. Figure 1 retains all 17 SNR settings; the supplied plot-value file also records every zero rate. The high-SNR errors of the sparse/EM method are not removed from the reported totals.

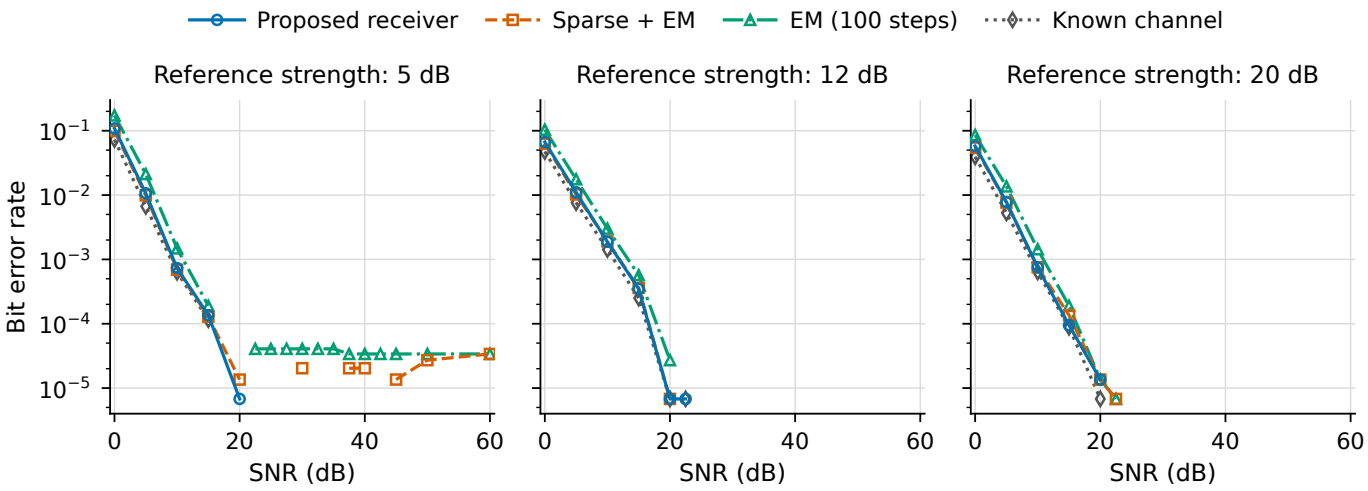


Fig. 1. Bit error rate from 0 to 60 dB, averaged over the three path settings. Every method uses the same data and detector. "Known channel" gives the detector the true channel. Zero observed rates are omitted from the log scale and break the lines. They do not prove a zero population error rate. This plot includes the higher-SNR sparse/EM errors outside the main paper's focused view.

X. Guide to the Historical Studies

The following retained studies explain the classical structure, original controller, alternative training and earlier failure mechanisms. Their uses of "our estimator," "main method," "primary" or "final" denote each local historical experiment. The original 21-context checkpoints and re-corruption variants are not the matched-30-context components of the current receiver. Historical runtime backends and statistical sampling designs remain as originally stated and are not substituted into the new final comparison. The main manuscript and the preceding sections define the current receiver and claims.

XI. Historical: Focused Compact-Controller Comparison

This follow-up tests whether the 188-parameter controller adds value over smaller learned controls. All four arms use spectral initialization and ten relaxed Rician EM stages. The step-only arm learns ten update parameters. The 30-parameter arm adds two filtering parameters per stage. The original 188-parameter arm uses its shared 8–16–2 controller. The new recurrent arm uses the same smooth Hankel filter but a different controller. We reuse the original selected checkpoints for the first three arms and train only the recurrent arm. This is a comparison of fixed designs, not a search for the best compact network.

A. Residual-Driven Recurrent Control

Let G^{t-1} be the current estimate, G^t_{EM} its exact EM update, and

$$G^t_{\mathrm{dc}} = G^{t-1} + \gamma_t (G^t_{\mathrm{EM}} - G^{t-1}), \qquad \gamma_t = 2\,\mathrm{sigmoid}(a_t). \tag{9}$$

The first controller input is shared by users within the received block; the second is specific to user k:

$$r_t = \log\left(1 + \frac{\|Z - |G^t_{\mathrm{dc}} S + B|\|_F^2}{NPv}\right), \tag{10}$$

$$u_{tk} = \log\left(1 + \frac{\|g^t_{\mathrm{EM},k} - g^{t-1}_k\|_2^2/N}{\max\{\|g^t_{\mathrm{dc},k}\|_2^2/N, 10^{-12}\}}\right). \tag{11}$$

Both inputs use only observations and the current estimate. A shared LSTM cell takes (r_t, u_{tk}) and has four hidden coordinates. Its hidden and memory states start at zero for every block and every user. States pass between stages, never between blocks or users. A linear map of the hidden state gives two outputs $(o_{tk,1}, o_{tk,2})$, with

$$\tau_{\mathrm{rel},tk} = 0.001 + \mathrm{softplus}(o_{tk,1}), \qquad \beta_{tk} = \mathrm{sigmoid}(o_{tk,2}). \tag{12}$$

The smooth singular-value rule is $s \mapsto s^3/(s^2+\tau^2)$, with $\tau = \tau_{\mathrm{rel},tk}\|\mathcal{H}(g^t_{\mathrm{dc},k})\|_F/\sqrt{q}$. Antidiagonal averaging gives $g^t_{\mathrm{str},k}$, and the next estimate is $g^t_k = g^t_{\mathrm{dc},k} + \beta_{tk}(g^t_{\mathrm{str},k} - g^t_{\mathrm{dc},k})$. The LSTM has 128 parameters, its output map has ten, and the ten EM steps give 148 in total. Initial relative strength and mixing are both 0.1; initial γ_t is one.

NA-ALISTA uses observable residual and backprojected-residual summaries to control linear sparse recovery [2]. Our adaptation instead controls a Hankel filter inside Rician EM. It uses the normalized inputs above, zero initial recurrent states, hidden-state readout, and softplus/sigmoid outputs. It learns filtering strength and mixing, not NA-ALISTA's target-dependent update step. The ten EM steps remain learned stage constants. Its loss uses withheld magnitudes rather than true signals. Results therefore concern this adaptation, not reproduction or superiority over published NA-ALISTA. Input features and architecture both differ from the 188-parameter arm, so this test alone does not isolate which difference causes a gain.

B. Matched Training and Fresh Testing

Each learned arm uses 16,384 observation-only training blocks, 2,048 separate validation blocks, 20 epochs, Adam with learning rate 0.001, batch size 32, and seeds 9101–9103. Training uses single precision. For zero-based epoch e and training seed s, the mask seed is $s + 10000 + e$ and the batch-permutation seed is $s + 20000 + e$. These match the original arms. Each mask gives 21 fitting and nine withheld pilots. Validation masks use seed 2026092203. Checkpoint selection includes epoch zero and minimizes the equal-weight mean held-out Rician loss over the 126 validation strata. Stratum labels balance offline selection; they are not controller inputs. No true channels select checkpoints. All three seeds are retained. They share one training dataset, not three independent training populations.

Source and development-data hashes are frozen before training. Selected checkpoint hashes are frozen before new test observations are generated. Test-bank master seeds are 20261012110001 and 20261012120001. In each bank, cell c uses world indices $16c$ through $16c + 15$. Trial IDs retain both master seed and index. The 126 primary cells cross SNR -10 to 20 dB in 5 dB steps, RSR $5, 12, 20$ dB, and six path regimes: independently drawn 3–7 paths per user, or fixed $2, 4, 8, 12, 16$ paths. Here $N = 32$, $K = 3$ and $P = 30$. Two extra cells test RSR 0 dB at SNR 5 dB and SNR 60 dB at RSR 12 dB; they are excluded from the primary average. Both use random 3–7 paths. The total is 4,096 fresh paired blocks. Every method receives the same channel, pilots, reference and noise realization. Inference uses all 30 pilots, retaining the original change from 21 fitting pilots during training. This follow-up does not resolve that change separately.

Offline-tuned OptShrink and per-block in-loop rank selection provide classical context. Their earlier validation-selected settings remain frozen; no new classical tuning uses these tests.

C. Reporting, Cost and Verification

For each cell and trained model, we calculate dB NMSE from summed squared channel error divided by summed true-channel energy across both banks. The primary statistic averages equally over 126 cells and then over three training seeds. The three predefined contrasts compare the 188-parameter arm with the step-only, 30-parameter and recurrent arms. We use 4,000 paired bootstrap resamples within each bank and cell, retaining pairing across all methods and seeds. Two-sided percentile intervals use tail probability $0.05/(2 \cdot 3)$ for a nominal 95% Bonferroni familywise level. They are

TABLE V
FRESH MATCHED COMPACT COMPARISON: 4,032 PRIMARY WORLDS AND 64 SEPARATE STRESS WORLDS. LOWER NMSE IS BETTER. NEURAL SCORES AVERAGE THREE MODELS, NOT AN ENSEMBLE. ALL TIMINGS INCLUDE INITIALIZATION, USE ONE PYTORCH CPU THREAD AND COUNT RANK SEARCH/REFITTING.

Method	Weights	NMSE (dB)	Time (ms)
Learned steps only	10	-9.773	1.79
Global filtering	30	-11.857	3.41
Spectral/state controller	188	-13.692	4.87
Recurrent control	148	-11.725	4.15
OptShrink, offline	—	-13.204	15.81
OptShrink, block rank	—	-14.144	110.56

TABLE VI
188-PARAMETER GAINS: POSITIVE FAVORS THE SPECTRAL/STATE CONTROLLER. THREE PREDECLARED COMPARISONS WITH BONFERRONI-ADJUSTED PAIRED INTERVALS. WORST-CELL HARM IS DESCRIPTIVE, NOT AN UPPER CONFIDENCE BOUND.

Comparator	Gain (dB)	95% interval	Worst harm
Learned steps only	+3.919	[+3.789, +4.001]	2.735
Global filtering	+1.835	[+1.805, +1.871]	0.580
Recurrent control	+1.967	[+1.932, +2.006]	0.176

conditional on the fitted models and development data; they do not measure uncertainty from new training datasets. Classical comparisons are descriptive in this follow-up. We also report per-cell results, seed and bank variation, worst-cell deterioration, pooled NMSE, and fractions with NMSE above $0.5, 1, 2$.

Timing uses single-thread PyTorch CPU, double precision and batch size one after training has ended. One independent block per cell uses seed 20261012130001. After warmup, each method runs three times in randomized order. Times include initialization, controller features, filtering, rank search and refitting, but exclude model loading and file access for every method. Training cost is separate. These are measured implementation times, not hardware-independent complexity claims.

Verification checks parameter counts, constant-control equivalence, finite outputs and gradients, a central-difference gradient, single/double precision agreement, state reset, user permutation, and deterministic optimizer resumption. The result audit regenerates all test inputs, checks hashes and trial IDs, replays one prediction per model and cell/bank, compares classical predictions with an independent NumPy implementation every eighth cell, and recalculates headline statistics with scalar sums. No outcome is inferred from these checks before the complete result audit.

D. Full Compact-Control Results

The audit regenerated all 4,096 test worlds, replayed 3,072 representative neural predictions and independently replayed 64 classical predictions in NumPy. Maximum absolute replay difference was $4.40e-13$. Headline scores were also recomputed with scalar loops. These numerical checks do not establish statistical optimality or global recovery.

TABLE VII
DIFFERENT ERROR SUMMARIES ON THE SAME PRIMARY WORLDS. POOLED NMSE POOLS LINEAR ERRORS BEFORE TAKING DB AND IS NOT THE PRIMARY SCORE. THRESHOLD RATES ARE PERCENTAGES.

Method	Pooled (dB)	$> .5$	> 1	> 2
Learned steps only	-1.651	29.99	22.84	14.24
Global filtering	-2.438	27.98	19.56	12.52
Spectral/state controller	-6.664	19.43	3.10	0.00
Recurrent control	-2.396	28.04	19.78	12.62
OptShrink, offline	-6.564	19.12	4.46	0.00
OptShrink, block rank	-6.228	19.27	6.77	0.27

TABLE VIII
SEPARATE STRESS TESTS, 32 PAIRED WORLDS EACH. WEAK REFERENCE: SNR 5/RSR 0 DB. NEARLY NOISELESS: SNR 60/RSR 12 DB. NEITHER ENTERS SELECTION OR THE PRIMARY AVERAGE.

Method	Weak reference	Nearly noiseless
Learned steps only	-4.065	-44.417
Global filtering	-6.505	-30.975
Spectral/state controller	-6.842	-6.321
Recurrent control	-5.623	-20.101
OptShrink, offline	-10.823	-67.701
OptShrink, block rank	-12.625	-70.336

TABLE IX
PRIMARY SCORES BY TRAINING INITIALIZATION. THESE SEEDS SHARE THE SAME TRAINING DATA AND DO NOT REPRESENT INDEPENDENT TRAINING POPULATIONS.

Method	9101	9102	9103
Learned steps only	-9.770	-9.774	-9.774
Global filtering	-11.852	-11.853	-11.865
Spectral/state controller	-13.784	-13.449	-13.843
Recurrent control	-11.707	-11.740	-11.728

TABLE X
RECURRENT TRAINING RECORDS. TIME INCLUDES VALIDATION AND PARALLEL-WORKER CONTENTION; IT IS NOT AN ISOLATED-DEVICE TRAINING BENCHMARK.

Seed	Selected epoch	Held-out NLL	Minutes
9101	20	3.43585	17.37
9102	19	3.43353	17.09
9103	20	3.43426	17.34

XII. HISTORICAL: TRAINING-RECIPE CHECKS AND NOISE-INPUT SENSITIVITY

The original 188-parameter estimator remains the proposed method. The checks below change its training recipe or select between two frozen models; they do not add a new estimator architecture. Each study has its own fresh test cases. Their numerical averages must not be pooled as if they came from one experiment. A severe error means per-case channel NMSE > 1, not a measured communication outage or numerical divergence.

A. All-Pilot Rician Training

The original loss estimates a channel from 21 pilot columns and predicts nine held-out columns. Inference uses all 30 columns. To test this change in input geometry, we also

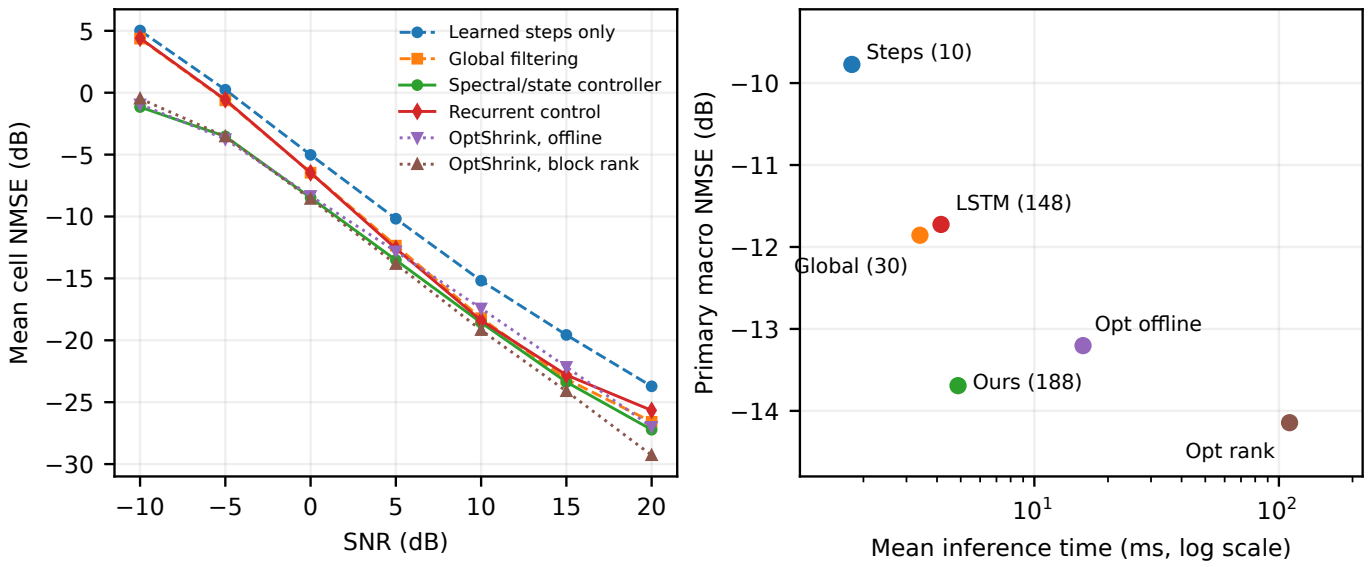


Fig. 2. Fresh compact-control comparison. Left: each SNR point averages 18 reference/path cells. Right: primary accuracy and measured serial inference time. These descriptive summaries do not imply cellwise confidence guarantees.

trained using all pilot columns, following the re-corruption principle [3]. For an observed magnitude $z = |\mu + w|$, with noiseless mean $\mu = (GS + B)_{np}$ and $w \sim \mathcal{CN}(0, v)$, draw an independent $\eta \sim \mathcal{CN}(0, v)$ and fix $\alpha > 0$. Form

$$z_{\text{in}} = |z + \alpha\eta|, \qquad z_{\text{target}} = |z - \eta/\alpha|. \tag{13}$$

The input and target complex-noise variances are $v(1 + \alpha^2)$ and $v(1 + \alpha^{-2})$, respectively. The EM update uses the first variance, and the predictive Rician loss uses the second.

This construction does not require the unobserved complex field. To check its distribution, write $Y = \mu + w$ and rotate η by $\arg Y$. Circular symmetry leaves the rotated noise independent of Y and unchanged in distribution. The resulting complex noises are $w + \alpha\eta'$ and $w - \eta'/\alpha$. They are jointly proper Gaussian, with cross-covariance $v - v = 0$. Their magnitudes are therefore independent conditional on μ. They are *not* independent conditional on a fixed saved magnitude z. Repeated artificial corruptions of one block are augmentations, not new channels. This argument assumes known, independent circular Gaussian field noise and fixed α; it is not an unbiased-channel-NMSE result. The predictive likelihood and channel error remain different objectives.

An initial development screen used 2,304 training blocks, 384 validation blocks, two initialization seeds (10401 and 10402), and ten epochs. Its six primary cells had SNR $-5, 5, 15$ dB, RSR $5, 12$ dB, and random 3–7 paths per user. The 384 primary test worlds and 128 separate stress worlds were generated after model freezing. Fixed-final-epoch scores were primary: original, $\alpha = 0.5$, and $\alpha = 1$ gave -12.392, -12.835, and -13.365 dB, with severe rates 1.562%, 0.521%, and 0.391%. The two seed gains for $\alpha = 0.5$ were -0.222 and 1.109 dB; for $\alpha = 1$ they were 0.747 and 1.198 dB. Thus the screen supported testing $\alpha = 1$, not a universal gain or reduced failure rate. Checkpoints selected by the common 21/9 validation loss were secondary in this small screen.

B. Matched Full-Budget Comparison

The larger study fixed $\alpha = 1$ before generating new test data. It used the original 16,384 training blocks and 2,048 validation blocks, three matched initialization seeds (9101–9103), 20 epochs, Adam with step size 10^{-3}, and batches of 32. Both learned recipes retained the same 188 parameters and ten EM/filtering stages. New artificial noise was drawn each epoch. Initialization and epoch data orders matched the original runs. Matching examples and optimizer updates does not match training operations, noise distributions, or objectives. The experiment tests the whole training recipe, not pilot geometry alone; 20 epochs do not establish convergence.

Both methods' primary checkpoints minimize equal-condition 21/9 validation Rician loss, with epoch zero eligible. This preserves the original selection rule but may favor its training recipe. Final-epoch comparisons are reported separately. The full-budget study contains 8,192 fresh worlds in two banks (seeds 20261014211001 and 20261014221001), with 32 cases per cell and bank. Its 126 primary conditions cross SNR -10 to 20 dB in 5 dB steps, RSR $5, 12, 20$ dB, and random 3–7 or fixed $2, 4, 8, 12, 16$ paths per user. Here $N = 32$, $K = 3$, and $P = 30$. Two stress cells add 128 worlds but do not enter the primary metric. Offline and per-block rank-selected OptShrink reuse frozen validation choices; per-block selection scores ranks 0–15 on 21/9 columns and refits all 30 columns.

The primary score pools the two banks' squared errors and channel powers within each cell, takes their dB ratio, averages the 126 cells equally, then averages the three model-specific scores. This is neither an ensemble nor the dB of one pooled full-range ratio. All-pilot training improves this score by 0.277 dB over the original, but is 0.171 dB worse than block-rank OptShrink. It wins 106/126 cells against the original, with a largest observed loss of 0.527 dB. Severe errors increase from 2.819% to 4.473%; hence the average improvement is

not a general reliability improvement. The final-epoch gain is 0.242 dB. All-pilot training reduces the nearly noiseless error floor but remains far behind OptShrink there, and does not fix weak-reference performance. Tables XI–XIII give the sensitivity, uncertainty and training records.

The two declared average-gain intervals use 4,000 paired resamples within bank/cell, with Bonferroni adjustment. Pairing is retained across methods and model seeds. The intervals condition on the fitted models and shared training dataset; three initializations are not three independent training populations. CPU timings use one PyTorch thread, float64, batch one, and three randomized repeats per condition. They include initialization and rank search/refitting, but exclude loading, file access, and offline fitting. Original and all-pilot inference take about 4.81 ms, compared with 15.66 ms for offline OptShrink and 108.08 ms for block-rank OptShrink. These are measured implementation costs, not a hardware-independent complexity claim.

C. One-Seed Mixed-Training Screen

A separate test used seed 9101 and the same full training budget, with exactly half the 512 batches per epoch assigned to each recipe. Each batch still used one model evaluation and one optimizer update. The two losses were averaged per measurement and logged separately; their different noise levels make their raw scales incomparable. The same 21/9 validation rule selected epoch 19. Testing used 4,096 new worlds (bank 20261015111001), including 4,032 primary and 64 stress worlds.

Mixed training improved the selected original by 0.152 dB. Its 0.033 dB deficit to selected all-pilot training is descriptive: the adjusted gain interval $[-0.0668, 0.0008]$ dB includes zero. Its severe rate was 3.745%, between the original's 2.902% and all-pilot's 4.762%. Compared with all-pilot, it corrected 42 severe outcomes and introduced one; compared with the original, it introduced 34 and corrected none. At -10 dB it reduced both mean-case error and severe frequency versus all-pilot. At -5 dB it reduced severe frequency but increased mean-case NMSE by 0.0100. Its 5.008 ms runtime was only 0.56% above the original's measured time. Table XIV retains all selected/final results and the matched classical controls.

The predefined screen required a 0.2 dB gain over the original, lower mean-case error and severe frequency than all-pilot at both -10 and -5 dB, an overall severe-rate increase of at most 0.2 percentage points over the original, and runtime at most 1.05 times the original. It passed the severe-frequency and runtime checks, but failed the gain, low-SNR mean-error, and overall severe-rate checks. This one-seed development result is not promoted to the main method. Its 2,000 paired resamples condition on that one fit; only the four macro-gain contrasts receive multiplicity adjustment. The screening tolerances are practical rules, not statistical noninferiority claims.

D. Simple Selection and Incorrect Noise Inputs

We next froze the seed-9101 original and all-pilot models without retraining. A scalar rule uses nominal SNR $10\log_{10}(3/\widetilde{v})$ and selects the original below a threshold and all-pilot above it. The constant 3 is the known expected signal power under this normalized three-user benchmark, not the unknown realized channel power. The rule is not established for arbitrary unknown path loss. Candidate cutoffs were $-7.5, -2.5, 2.5, 7.5, 12.5, 17.5$ dB, plus both constant policies. Selection used only the 2,048 validation observations, fixed 21/9 prediction loss, and equal condition weights, without channel labels.

Validation selected *always original*: average loss 3.372665, compared with 3.372740 for the 17.5 dB cutoff and 3.386371 for always all-pilot. We did not force a switch or revise the cutoff after testing. The 21/9 criterion may favor the original recipe and need not rank all-30-pilot channel accuracy identically. This negative result rejects this particular selection procedure, not every possible selector.

The new bank (20261015151001) has 4,224 worlds: 4,032 primary cases, 64 weak-reference/nearly-noiseless stress cases, and 128 off-grid cases at SNR ± 2.5 dB with RSR 5/12 dB. Each physical observation is reused with supplied variance $\widetilde{v} = v10^{\delta/10}$, $\delta \in \{0, -3, +3\}$ dB. Raw magnitudes, pilots, reference, actual noise, and true channels remain fixed. Every comparator and the router receive the same biased variance. The three settings therefore are not three independent test populations.

Table XV shows a substantial sensitivity to noise underestimation. The original's severe rate rises from 2.803% with correct variance to 11.111% at -3 dB; the all-pilot rate rises from 4.861% to 14.509%. Overestimation by 3 dB gives 3.398% and 5.432%, respectively. Because the frozen policy is constant, there are no route changes and its estimates equal the original's. The correct-noise 0.1 dB gain criterion fails, although the severe-rate, low-SNR and 1.05 runtime criteria pass. The selector stores 376 model weights plus the cutoff, evaluates only one 188-parameter model per case, and takes 4.945 ms versus 4.875 ms for the original. No OptShrink rerun is included in this screen. Its 2,000 paired intervals are descriptive, conditional on the two fitted models, and unadjusted.

E. Oracle Phase Diagnostic and Audit Scope

A separate diagnostic uses 576 new worlds (seed 20261014031001), with 64 per SNR/RSR cell. It evaluates the three frozen original models at SNR $-5, 5, 15$ dB and RSR $0, 5, 12$ dB. The raw, truth-aligned global-phase, and truth-aligned antenna-row-phase average scores are -10.833, -10.912, and -12.853 dB. The much larger row-wise change at weak reference identifies an error component, not an implementable receiver gain. Alignment can remove noise as well as phase error; it does not prove a distinct local minimum or a physical reference-drift mechanism. No learned setting is chosen from this diagnostic.

Saved audits checked paired identifiers, source/data/model hashes, complete histories, frozen validation choices, finite outputs, and representative prediction replays. The full-budget study replayed 384 neural and 64 independent classical predictions; mixed training replayed 60 neural and 20 classical

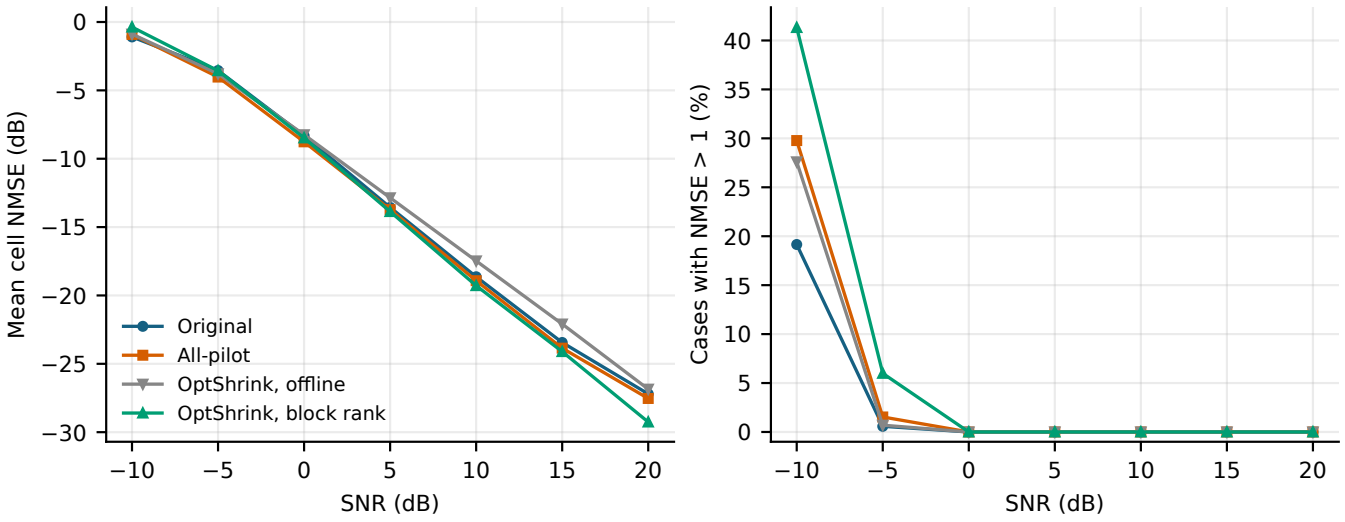


Fig. 3. The separate full-budget training comparison. Each SNR averages 18 reference/path conditions. Better average channel accuracy does not imply fewer severe errors. Curves use the same 8,064 primary worlds; neural scores average three fits, not an ensemble.

predictions; the selector replayed 1,188 predictions. Maximum neural discrepancies were below 5×10^{-15}. This is not a replay of every training update. The manuscript material generator independently recomputes primary NMSE and severe rates from the saved error arrays. Small evidence files retain all condition-level summaries, protocols, phase diagnostics, selection scores and input hashes. These checks establish consistency of the numerical record, not hardware validation, optimality, convergence, or robustness to new training populations.

XIII. Historical: Broad OptShrink Comparisons: Full Numerical Record

Study A tunes OptShrink on the original 2,048 validation observations and masks. Study B inherits that lookup and selects rank from each new block of observations. Neither study retrains the compact estimator. Each has two independent test banks, 126 primary cells and two separate stress cells. All 30 pilots are reused for the final fit, without additional observations. The inference lookup uses supplied noise and reference strength, not true channel power or path count.

Study A uses test-bank seeds 202610210001 and 202610310001; Study B uses 202610610001 and 202610710001. Each cell adds 100 times its zero-based index to the bank seed. Both use 16 worlds per cell and bank. The selected-model seeds are 9101, 9102 and 9103. Sources, settings and checkpoints are hashed before test generation. Settings remain frozen after the test data are inspected.

The primary score averages 126 per-cell dB ratio-of-sums NMSE values, after pooling the two equal-sized test banks within each cell. It is not the dB of one pooled error ratio. Learned scores average the three separate model scores. Bootstrap resampling preserves pairing within each bank/cell; confidence intervals condition on these weights and the offline validation choices. They do not cover repeated training or offline selection uncertainty.

A. Declared Contrasts and All Comparators

The numerical tables separate the declared average-dB contrasts, alternative error summaries, individual models, test banks and stress cases. These summaries are not interchangeable: a gain in one average does not establish a gain in every cell or at every error threshold. All individual-cell scores are retained in the evidence records.

B. Frozen Settings and Numerical Verification

The online selector searches a shared rank, not separate ranks for each user. Its mixing strength and iteration count are inherited from the matching in-loop or final-only row in the offline lookup. Thus the rank comparison does not exhaust all possible classical policies. Candidate initialization uses only the first 21 pilots. After scoring the other nine, the winner starts again from all 30. Joint selection evaluates both placements and counts both searches in timing. Search work shared while generating several experimental arms is not used to understate standalone timing.

Study A: all 4,096 test worlds regenerated exactly; 3,584 representative predictions replayed. Maximum recorded independent classical absolute output difference was 5.73×10^{-13}. An independent scalar-loop calculation reproduced every primary score. These checks test the saved implementation and data, not an external reproduction of the study.

Study B: all 4,096 test worlds regenerated exactly; 2,816 representative predictions replayed. Maximum recorded independent rank-selector absolute output difference was 1.26×10^{-12}. An independent scalar-loop calculation reproduced every primary score. These checks test the saved implementation and data, not an external reproduction of the study.

Study A also regenerated all 2,048 validation observations and replayed selection from saved scores. Study B independently checked both rank-score vectors, chosen settings and all-pilot refits with NumPy/SciPy. Its new audit did not independently replay HS-Rician again; that backend check

TABLE XI

FULL-BUDGET STUDY: SELECTED AND FINAL-EPOCH SENSITIVITY, PLUS SEPARATE STRESS CELLS. PRIMARY SCORES USE 8,064 WORLDS; EACH STRESS CELL HAS 64 WORLDS. WEAK REFERENCE: SNR 5/RSR 0 DB. NEARLY NOISELESS: SNR 60/RSR 12 DB. STRESS SCORES DO NOT ENTER THE PRIMARY SCORE.

Method	Macro (dB)	Pooled (dB)	Severe (%)	Weak reference	Nearly noiseless
Original (selected)	-13.684	-6.650	2.82	-7.244	-6.949
All-pilot (selected)	-13.962	-6.711	4.47	-6.903	-15.240
Original (final)	-13.762	-6.684	3.04	-7.280	-6.921
All-pilot (final)	-14.005	-6.695	4.61	-6.865	-23.116
OptShrink, block rank	-14.132	-6.201	6.76	-11.257	-69.517
OptShrink, offline	-13.168	-6.568	4.04	-11.866	-66.684

TABLE XII

ALL-PILOT GAINS: POSITIVE FAVORS ALL-PILOT TRAINING. TWO PREDECLARED CONTRASTS, BONFERRONI-ADJUSTED CONDITIONAL 95% INTERVALS. WORST OBSERVED HARM IS NOT A CONFIDENCE BOUND.

Comparator	Gain (dB)	95% interval	Cell wins	Worst harm
Original (selected)	+0.277	[+0.263, +0.294]	106/126	0.527
OptShrink, block rank	-0.171	[-0.221, -0.117]	72/126	3.744

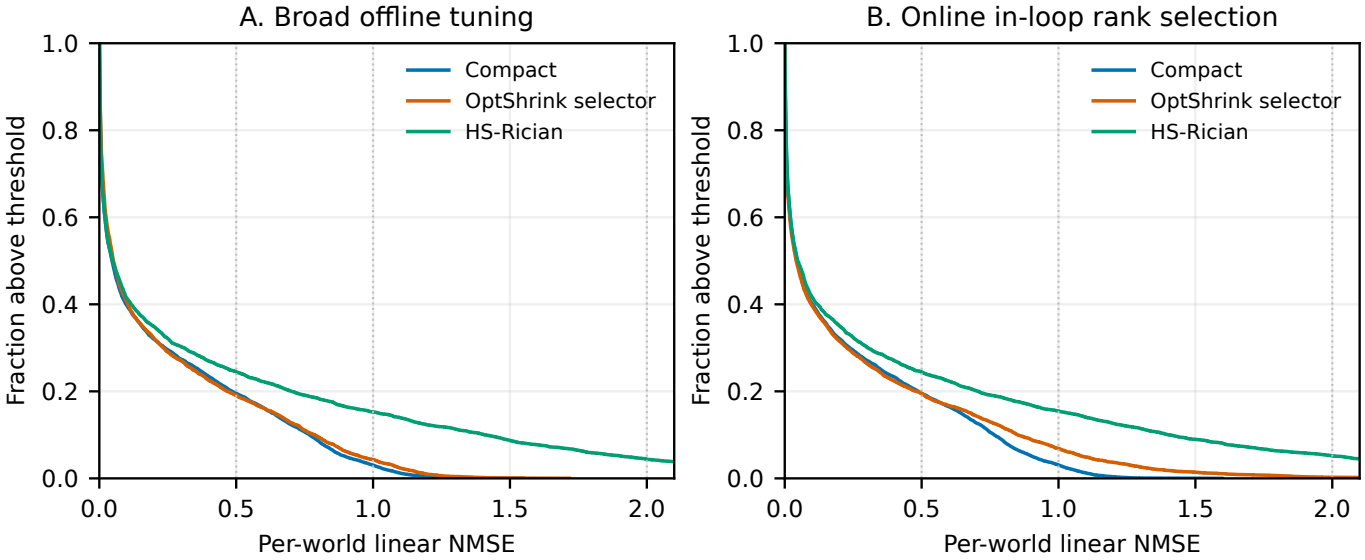


Fig. 4. Primary empirical exceedance curves over the displayed threshold range, separately by study. Learned curves average the three model-specific empirical distributions on the same worlds; predictions are not independent channels. All values, including larger errors, are retained in the evidence arrays. A lower rate at one threshold does not imply a lower rate at every threshold.

TABLE XIII

FULL-BUDGET TRAINING RECORDS. SEEDS SHARE ONE TRAINING DATASET. ALL-PILOT TRAINING TIMES INCLUDE VALIDATION AND CONCURRENT-WORKER CONTENTION; THEY ARE NOT ISOLATED-DEVICE BENCHMARKS.

Seed	Original	All-pilot	Best epoch	Minutes
9101	-13.786	-13.978	16	29.05
9102	-13.446	-13.938	18	28.99
9103	-13.822	-13.969	17	29.31

is in Study A. Both studies passed finite-output, source-hash, checkpoint-hash and freeze-before-generation checks. No failed trial was removed. Full per-cell values and source records are supplied as JSON evidence in the manuscript archive; full observation and prediction arrays remain in the experimental archive.

XIV. HISTORICAL: RETAINED NEGATIVE FOLLOW-UPS

These studies tested possible improvements before the broad comparisons. None changed the three original 188-parameter checkpoints. Their outcomes are kept separate because they use different test banks and budgets.

A. A Controller Around OptShrink

The first hybrid used a learned mixing rule around OptShrink. On 704 fresh worlds, its seven-SNR average was -14.129 dB, compared with -14.535 dB for the stronger OptShrink control. It therefore lost 0.406 dB. Its ten-stage schedule and some ranks did not match the strong baseline. A clipped gate also had zero local derivative in many high-SNR cases. These were reasons to repair the comparison, not evidence that repairing it would produce an improvement.

TABLE XIV
SEPARATE ONE-SEED MIXED-TRAINING SCREEN: 4,032 PRIMARY WORLDS, NOT THE FULL-BUDGET TEST BANK. MEAN-CASE NMSE IS LINEAR. SELECTED-CHECKPOINT COMPARISONS ARE PRIMARY; FINAL CHECKPOINTS ARE SENSITIVITY CHECKS.

Method	Macro (dB)	Case mean	Severe (%)	Time (ms)
Original (selected)	-13.799	0.2222	2.90	4.98
All-pilot (selected)	-13.984	0.2200	4.76	4.96
Mixed (selected)	-13.951	0.2163	3.75	5.01
Original (final)	-13.672	0.2224	3.00	4.98
All-pilot (final)	-14.004	0.2209	4.96	4.96
Mixed (final)	-13.966	0.2160	3.84	5.01
OptShrink, offline	-13.119	0.2257	4.69	16.06
OptShrink, block rank	-14.147	0.2454	6.70	115.20

TABLE XV
SEPARATE ONE-SEED SELECTOR/NOISE-INPUT SCREEN: THE SAME 4,032 PRIMARY WORLDS AT ALL THREE SUPPLIED-VARIANCE OFFSETS. PHYSICAL OBSERVATIONS AND ACTUAL NOISE ARE UNCHANGED. THE SELECTED RULE ALWAYS RUNS THE ORIGINAL MODEL.

Offset (dB)	Method	Macro (dB)	Case mean	Severe (%)	Time (ms)
+0	Original (selected)	-13.817	0.2228	2.80	4.88
+0	All-pilot (selected)	-14.033	0.2207	4.86	4.86
+0	Selected rule	-13.817	0.2228	2.80	4.95
-3	Original (selected)	-13.195	0.4064	11.11	4.85
-3	All-pilot (selected)	-13.007	0.4962	14.51	4.85
-3	Selected rule	-13.195	0.4064	11.11	4.94
+3	Original (selected)	-13.228	0.2652	3.40	4.86
+3	All-pilot (selected)	-13.781	0.2448	5.43	4.88
+3	Selected rule	-13.228	0.2652	3.40	4.94

TABLE XVI
SEPARATE PHASE DIAGNOSTIC: 64 WORLDS PER CELL, AVERAGED ACROSS THREE ORIGINAL FITTED MODELS. BOTH ALIGNED SCORES USE THE TRUE CHANNEL AND ARE ORACLE-ONLY, NOT RECEIVER PERFORMANCE. LAST COLUMN IS THE FRACTION OF SQUARED ERROR REMOVED BY ROW-WISE ALIGNMENT.

SNR	RSR	Raw (dB)	Global oracle	Row oracle	Removed (%)
-5	0	0.366	0.296	-2.176	44.2
-5	5	-2.184	-2.263	-2.932	15.8
-5	12	-4.288	-4.331	-4.820	11.5
5	0	-7.110	-7.270	-11.051	59.6
5	5	-13.761	-13.834	-14.705	19.5
5	12	-14.012	-14.071	-14.786	16.3
15	0	-8.306	-8.386	-15.192	79.5
15	5	-23.972	-24.078	-25.005	21.2
15	12	-24.226	-24.269	-25.007	16.5

TABLE XVII
LEARNED GAIN (POSITIVE FAVORS LEARNING), THREE-CONTRAST-ADJUSTED 95% INTERVALS WITHIN EACH STUDY, AND LARGEST OBSERVED PRIMARY-CELL DETERIORATION. CELLWISE SAFETY IS NOT ESTABLISHED.

Study	Comparator	Gain (dB)	Adjusted interval	Worst harm (dB)
A	Offline OptShrink, selected placement	0.537	[0.450, 0.695]	3.117
A	Offline OptShrink, in-loop	0.565	[0.477, 0.723]	3.117
A	Offline OptShrink, final-only	0.738	[0.621, 0.925]	3.686
B	Online rank + placement selection	-0.449	[-0.569, -0.358]	4.478
B	Online rank, in-loop	-0.507	[-0.562, -0.437]	5.495
B	Online rank, final-only	-0.076	[-0.226, 0.025]	3.445

The corrected pilot used the strong baseline's ranks and noise-dependent iteration schedule, removed the hard clipping, and trained two initializations on one shared dataset. Settings were fixed before 704 new test worlds were generated. The learned gate started at 0.95 where the baseline used strength one; a near-initial control accounted for this difference. Table XXVI gives the primary seven-SNR averages. Four stress cells were reported separately.

The corrected hybrid gained only 0.019 dB over the strong starting baseline, with conditional pointwise 95% interval $[0.004, 0.033]$ dB. It lost 0.022 dB to retuned classical mixing and 0.040 dB to the simpler stage-only learned rule. Mean same-backend processing times were about 7.2 ms for the hybrid and 6.2 ms for OptShrink. The corrected gate had no exact-zero local slopes in the recorded tests, but this did not make it the better estimator. The screen failed, and the

TABLE XVIII
STUDY A: ALL PRIMARY AVERAGES AND MEAN END-TO-END CPU TIME. FIXED-BUDGET LABELS DENOTE EM UPDATES, NOT MATCHED RUNTIME.

Method	NMSE (dB)	Time (ms)
Offline OptShrink, selected placement	-13.028	15.61
Offline OptShrink, in-loop	-13.000	15.57
Offline OptShrink, final-only	-12.827	6.16
OptShrink, 5 updates	-9.816	2.47
OptShrink, 10 updates	-12.336	4.56
OptShrink, 20 updates	-12.926	8.73
OptShrink, 50 updates	-12.941	21.20
OptShrink, 100 updates	-12.951	41.88
Original narrow OptShrink lookup	-11.880	6.31
EM-GS, 100 updates	-9.922	13.08
HS-Rician, one sweep	-12.996	98.91
Compact estimator (188 weights)	-13.565	4.90

TABLE XIX
STUDY B: ALL PRIMARY AVERAGES AND MEAN END-TO-END CPU TIME. FIXED-BUDGET LABELS DENOTE EM UPDATES, NOT MATCHED RUNTIME.

Method	NMSE (dB)	Time (ms)
Online rank, in-loop	-14.154	108.70
Online rank, final-only	-13.723	13.96
Online rank + placement selection	-14.096	114.17
Offline OptShrink, selected placement	-13.196	15.76
Offline OptShrink, in-loop	-13.151	15.68
Offline OptShrink, final-only	-12.863	6.23
Original narrow OptShrink lookup	-12.041	6.38
EM-GS, 100 updates	-9.958	13.15
HS-Rician, one sweep	-13.059	99.88
Compact estimator (188 weights)	-13.647	4.96

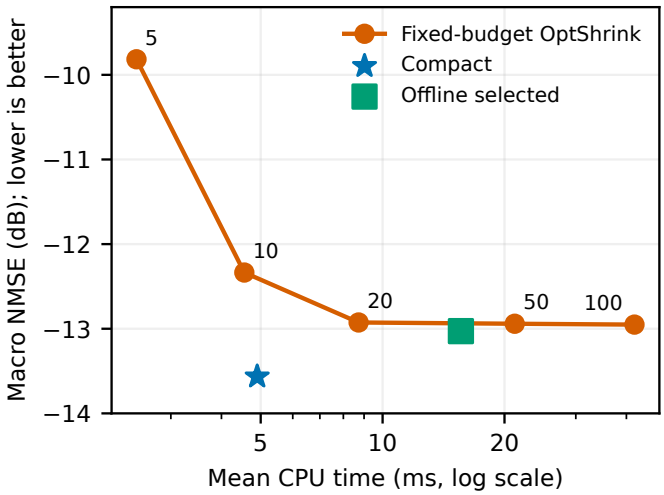


Fig. 5. Study A accuracy–cost curve. Labels are fixed update budgets, each tuned separately for placement, rank and mixing. The selected lookup may use different budgets by noise/reference pair. No equal-runtime optimum is claimed.

hybrid was not expanded. Its numerical audit regenerated 2,720 worlds and checked 198 checkpoint predictions and 11 independent classical predictions.

B. Stopping, Filter Release and a Larger Learned Control

A separate 704-world pilot included two independent 352-world banks, each with 32 worlds in seven primary and four stress cells. The first bank tested observable stopping and releasing the structural filter. Frozen OptShrink, observed stopping and observed release reached -14.530, -14.498 and -14.449 dB, respectively. Release lost 0.050 dB to observed stopping and 0.081 dB to frozen OptShrink. Stopping improved the weak-reference diagnostic, but not the primary average. It computed all 100 updates before choosing an output; its selected iteration count was not its total processing cost.

The second bank compared a newly trained compact model with an independently implemented exact-EM Transformer with 1,585,920 parameters. Both used 4,096 training cases, 12 epochs, matching masks and order, one training seed, and held-out likelihood for selection. Selected NMSEs were -14.325 dB for the compact model, -9.726 dB for the Transformer, -14.289 dB for OptShrink and -10.116 dB for EM100. The compact–OptShrink gain interval, $[-0.095, 0.176]$ dB, included zero. The Transformer was worse than EM100, so this pilot did not provide a strong learned comparator. It was not a reproduction of published URformer and supports no superiority claim over that method. Neither training budget establishes convergence.

C. An Error-Model Feasibility Check

A further screen examined whether local sensitivity could predict the error entering a Hankel filter. It used 24 fixed channel/pilot/reference worlds, 256 repeated noise draws and eight separate prediction anchors per world. Noise repetitions were not new channel realizations. The ordinary-start covariance check passed six of ten primary groups, below its required eight; the total-error/bias check passed seven. At SNR -10 dB after 100 EM updates, predicted variance was 16.48 times observed variance. With a weak reference it was 5.72 times. Estimated bias accounted for about 57% of total error at 20 updates and 22% at 100 updates in the weak-reference group.

Two supplementary finite-difference checks exceeded their original tolerance. Smaller steps and independent automatic differentiation resolved the numerical discrepancy. Ordinary-start relative Jacobian errors in the two problematic settings were about 0.0113% and 0.0023%, too small to explain the main prediction failure. The scientific screen remained failed. This study concerned unfiltered, row-separable EM; it did not validate an error model for interleaved Hankel reconstruction or establish an optimal shrinker. No new filter was promoted on the basis of these predictions.

D. What These Checks Leave Open

An observable likelihood score is not a channel-error guarantee. A smooth gate is not enough to establish useful adaptation. More network parameters do not establish a stronger trained baseline. The retained failures narrow the claim to the tested original compact estimator; they do not supply evidence for a new hybrid, an uncertainty-aware filter or a general improvement to OptShrink.

TABLE XX
ALTERNATIVE SUMMARIES ON PRIMARY WORLDS. POOLED NMSE IS IN DB; PER-CASE MEAN NMSE IS LINEAR. LAST THREE COLUMNS ARE PERCENTAGES ABOVE THE FIXED THRESHOLDS. ZERO OBSERVED EXCEEDANCES DO NOT IMPLY ZERO POPULATION RISK.

Study	Method	Pooled	Case mean	> 0.5	> 1	> 2
A	Compact estimator (188 weights)	-6.704	0.221	19.54	3.03	0.00
A	Offline OptShrink, selected placement	-6.588	0.227	19.12	4.34	0.00
A	HS-Rician, one sweep	-4.216	0.399	24.48	15.23	4.37
A	EM-GS, 100 updates	-1.415	0.777	30.38	23.59	14.71
B	Compact estimator (188 weights)	-6.683	0.221	19.55	3.08	0.00
B	Online rank, in-loop	-6.217	0.247	19.52	6.87	0.22
B	HS-Rician, one sweep	-4.166	0.402	24.50	15.48	5.21
B	EM-GS, 100 updates	-1.392	0.778	30.63	23.26	14.96

TABLE XXI
TRAINING-SEED AND TEST-BANK SUMMARIES, KEPT DISTINCT. BANK DB SCORES NEED NOT AVERAGE TO THE SCORE OBTAINED AFTER POOLING THEIR LINEAR ERRORS.

Study	Subset	NMSE (dB)
A	Model 9101	-13.676
A	Model 9102	-13.327
A	Model 9103	-13.692
A	Compact estimator (188 weights), bank 1	-13.700
A	Compact estimator (188 weights), bank 2	-13.559
A	Offline OptShrink, selected placement, bank 1	-13.059
A	Offline OptShrink, selected placement, bank 2	-13.064
B	Model 9101	-13.759
B	Model 9102	-13.417
B	Model 9103	-13.765
B	Compact estimator (188 weights), bank 1	-13.666
B	Compact estimator (188 weights), bank 2	-13.648
B	Offline OptShrink, selected placement, bank 1	-13.207
B	Offline OptShrink, selected placement, bank 2	-13.183
B	Online rank + placement selection, bank 1	-14.196
B	Online rank + placement selection, bank 2	-14.050
B	Online rank, in-loop, bank 1	-14.166
B	Online rank, in-loop, bank 2	-14.149

TABLE XXII
STUDY A: DESCRIPTIVE MARGINAL AVERAGES OVER THE OTHER PRIMARY CONDITIONS. OPTSHRINK IS OFFLINE-SELECTED. THESE ARE NOT ADDITIONAL INDEPENDENT TESTS.

Study	Condition	Compact	OptShrink	Gain
A	SNR -10	-1.057	-0.835	0.222
A	SNR -5	-3.682	-3.894	-0.212
A	SNR 0	-8.346	-8.220	0.126
A	SNR 5	-13.481	-12.832	0.649
A	SNR 10	-18.636	-17.375	1.262
A	SNR 15	-23.521	-22.275	1.246
A	SNR 20	-26.232	-25.767	0.465
A	RSR 5	-12.494	-11.840	0.654
A	RSR 12	-13.968	-13.450	0.518
A	RSR 20	-14.234	-13.795	0.439
A	Paths 3–7	-14.295	-13.536	0.759
A	Paths 2	-15.863	-13.674	2.190
A	Paths 4	-14.704	-13.814	0.890
A	Paths 8	-13.099	-13.255	-0.156
A	Paths 12	-12.119	-12.309	-0.190
A	Paths 16	-11.310	-11.582	-0.272

TABLE XXIII
STUDY B: DESCRIPTIVE MARGINAL AVERAGES OVER THE OTHER PRIMARY CONDITIONS. OPTSHRINK IS ONLINE IN-LOOP RANK SELECTION. THESE ARE NOT ADDITIONAL INDEPENDENT TESTS.

Study	Condition	Compact	OptShrink	Gain
B	SNR -10	-1.073	-0.362	0.710
B	SNR -5	-3.605	-3.574	0.031
B	SNR 0	-8.284	-8.426	-0.142
B	SNR 5	-13.581	-13.888	-0.307
B	SNR 10	-18.694	-19.289	-0.595
B	SNR 15	-23.365	-24.210	-0.845
B	SNR 20	-26.926	-29.329	-2.403
B	RSR 5	-12.754	-13.098	-0.345
B	RSR 12	-13.985	-14.530	-0.544
B	RSR 20	-14.202	-14.834	-0.632
B	Paths 3–7	-14.342	-14.546	-0.203
B	Paths 2	-16.436	-17.151	-0.715
B	Paths 4	-14.653	-15.150	-0.498
B	Paths 8	-13.014	-13.495	-0.482
B	Paths 12	-12.210	-12.592	-0.382
B	Paths 16	-11.227	-11.990	-0.763

E. Retained Figures From the Earlier Studies

These plots retain the earlier narrow-population and operating-limit results. They use the protocols stated in their captions, not the new broad-study test banks.

XV. HISTORICAL: MATCHED ONE-SWEEP BASELINE AND TAIL-ERROR AUDIT

This post-review follow-up uses the already inspected 1,152-world OptShrink bank: 128 worlds in each of seven primary SNR cells and two stress cells. It generates no new channels and trains no new model. The protocol is frozen

TABLE XXIV
STRESS CONDITIONS, EXCLUDED FROM PRIMARY AVERAGES. WEAK REFERENCE: SNR 5/RSR 0 DB. NEARLY NOISELESS: SNR 60/RSR 12 DB. BOTH USE RANDOM THREE-TO-SEVEN-PATH CHANNELS.

Study	Condition	Method	NMSE (dB)
A	Weak reference	Compact estimator (188 weights)	-7.842
A	Weak reference	Offline OptShrink, selected placement	-11.905
A	Weak reference	EM-GS, 100 updates	-6.109
A	Nearly noiseless	Compact estimator (188 weights)	-6.693
A	Nearly noiseless	Offline OptShrink, selected placement	-66.886
A	Nearly noiseless	EM-GS, 100 updates	-65.604
B	Weak reference	Compact estimator (188 weights)	-7.787
B	Weak reference	Offline OptShrink, selected placement	-11.498
B	Weak reference	EM-GS, 100 updates	-6.261
B	Weak reference	Online rank + placement selection	-10.101
B	Weak reference	Online rank, in-loop	-10.101
B	Nearly noiseless	Compact estimator (188 weights)	-6.835
B	Nearly noiseless	Offline OptShrink, selected placement	-67.618
B	Nearly noiseless	EM-GS, 100 updates	-66.212
B	Nearly noiseless	Online rank + placement selection	-70.134
B	Nearly noiseless	Online rank, in-loop	-70.121

TABLE XXV
BROAD OFFLINE SETTINGS. WITHIN EACH PLACEMENT, COLUMNS GIVE RANK, MIXING FACTOR AND UPDATE COUNT. THE SELECTED COLUMN CHOOSES BETWEEN THOSE PLACEMENTS. RANK ZERO DENOTES NO FILTERING.

SNR / RSR	Loop r	β	T	Final r	β	T	Selected
-10 / 5	1	0.75	5	2	1.0	50	final
-10 / 12	1	0.5	5	1	0.75	5	final
-10 / 20	1	0.75	5	1	0.75	5	loop
-5 / 5	2	0.5	10	3	0.75	20	final
-5 / 12	2	0.5	5	2	0.75	5	loop
-5 / 20	3	0.75	5	3	0.75	5	loop
0 / 5	4	0.5	20	5	0.75	100	loop
0 / 12	4	0.5	10	5	0.75	10	loop
0 / 20	4	0.5	10	7	1.0	5	loop
5 / 5	7	0.5	100	8	1.0	100	loop
5 / 12	7	0.5	20	8	1.0	20	loop
5 / 20	8	0.75	10	8	1.0	10	loop
10 / 5	9	1.0	100	9	0.75	100	loop
10 / 12	10	0.75	20	9	1.0	50	loop
10 / 20	10	1.0	20	10	1.0	20	loop
15 / 5	9	0.5	50	10	1.0	100	loop
15 / 12	10	0.75	50	10	1.0	50	loop
15 / 20	10	0.75	50	10	1.0	50	loop
20 / 5	11	1.0	100	13	1.0	100	loop
20 / 12	12	1.0	50	12	1.0	50	loop
20 / 20	11	0.75	50	11	1.0	50	loop

TABLE XXVI
CORRECTED HYBRID PILOT: LOWER NMSE IS BETTER.

Method	NMSE (dB)
Strong OptShrink	-14.596
Retuned classical mixing	-14.637
Residual-based classical rule	-14.633
Stage-only learned mixing	-14.656
Corrected measurement-conditioned hybrid	-14.616

before running the added comparator, not before the historical bank was inspected. It must not be described as a new blinded confirmation. The seven primary cells contain 896 distinct worlds.

The one-sweep HS-Rician estimator fits every rank 1–16 on the first 21 pilots for 25 EM updates, scores Rician likelihood on the other nine, then refits the selected rank with 100 updates and all 30 pilots. Each candidate and refit uses one Cadzow sweep. The sweep count is the only changed setting relative to the saved HS-Rician baseline; the new one-sweep candidates select their own ranks. Rank 16 bypasses filtering. No true channel is an estimator input. Learned checkpoints, OptShrink settings, initialization and observation budgets remain unchanged.

The added one-sweep HS-Rician control reaches -13.314 dB, improving on four sweeps by 0.424 dB. Learning retains a 1.004 dB average gain, with paired interval $[0.922, 1.088]$ dB. This closes the one-versus-four-sweep gap on the same bank, not the comparison to every possible hard-rank method. In the separate weak-reference and two-path cells, one sweep instead beats learning by 1.976 and 1.144 dB.

At the original threshold $e_i/d_i > 1$, the seven-SNR reduction in failure rate versus in-loop OptShrink is 4.13 percentage

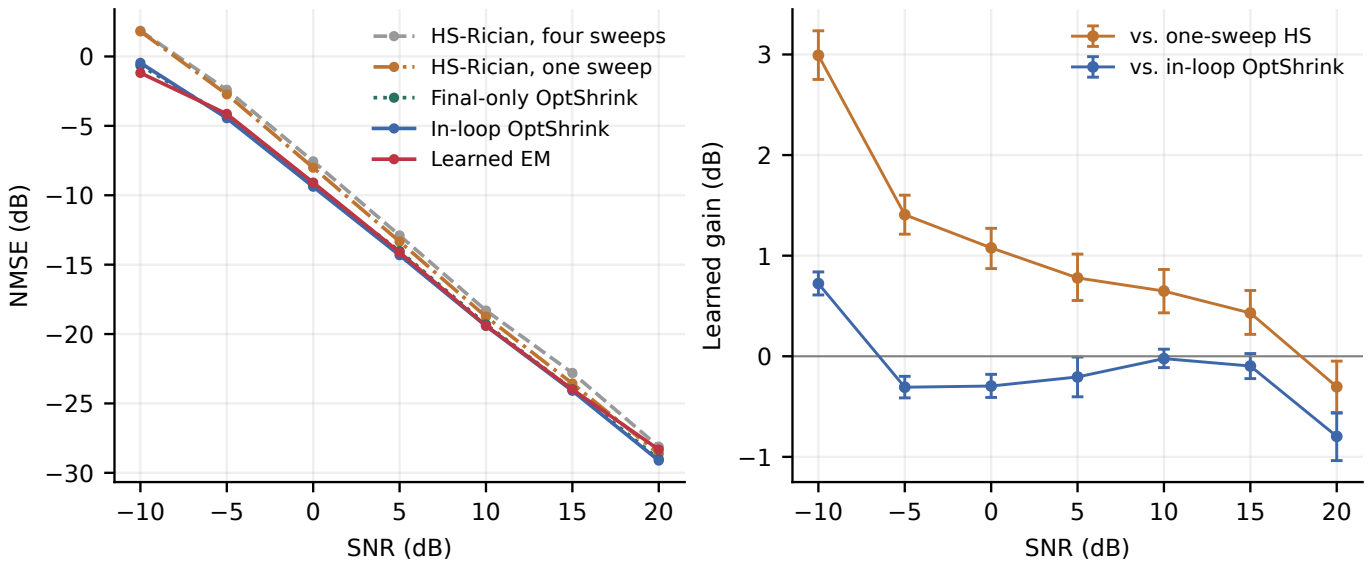


Fig. 6. Earlier narrow-population comparison: 128 paired trials per SNR and frozen learned models. The one-sweep HS-Rician result was added on the same saved worlds. Right: learned gains over one-sweep HS-Rician and in-loop OptShrink; positive favors learning. Intervals are pointwise and condition on the saved models.

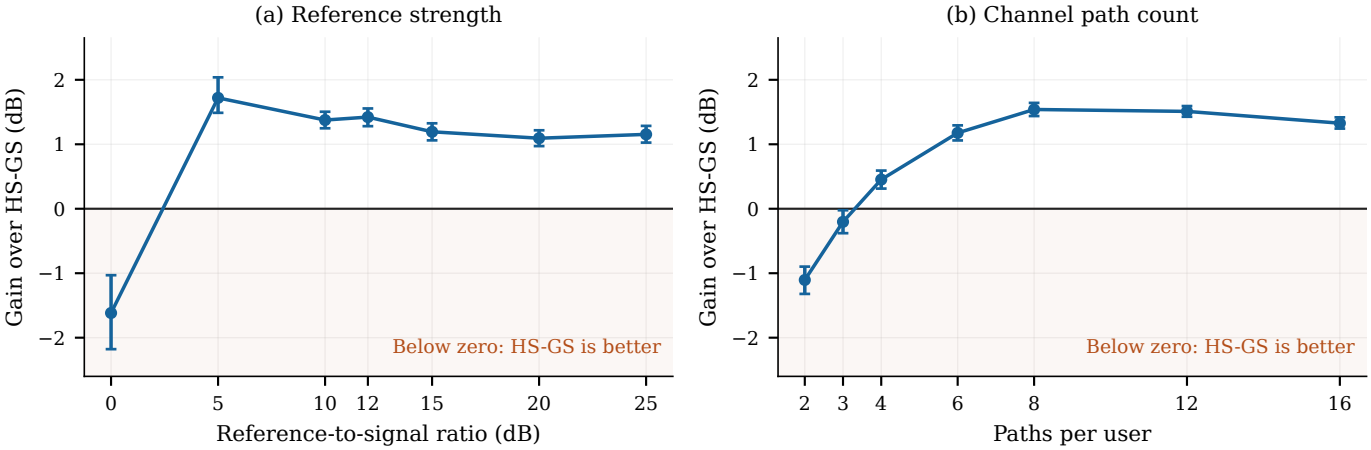


Fig. 7. Limits in the original learned study at SNR 5 dB. (a) Reference strength for three-to-seven-path channels. (b) Fixed path counts at RSR 12 dB. Gains compare adaptive EM with four-sweep HS-amplitude (HS-GS in the plot). Bars are conditional pointwise paired intervals; tested path counts overlap training.

TABLE XXVII
MATCHED SAVED-BANK RESULTS, LOWER DB NMSE IS BETTER. STRESS CELLS ARE EXCLUDED FROM THE SEVEN-SNR MACRO-AVERAGE. LEARNED VALUES AVERAGE THREE FIXED-MODEL DB SCORES, NOT ENSEMBLE ESTIMATES.

Cell	HS, one sweep	HS, four sweeps	In-loop OptShrink	Final OptShrink	Learned
SNR -10	1.808	1.850	-0.461	-0.650	-1.184
SNR -5	-2.731	-2.401	-4.445	-4.316	-4.138
SNR +0	-8.010	-7.558	-9.384	-9.333	-9.088
SNR +5	-13.335	-12.896	-14.320	-14.018	-14.114
SNR +10	-18.746	-18.306	-19.417	-19.306	-19.394
SNR +15	-23.561	-22.803	-24.088	-23.985	-23.991
SNR +20	-28.625	-28.118	-29.118	-29.010	-28.322
RSR 0 dB	-9.045	-8.307	-5.026	-6.613	-7.069
Two paths	-17.958	-17.833	-16.406	-16.592	-16.814
Seven-SNR mean	-13.314	-12.890	-14.462	-14.374	-14.319

points, with paired interval $[3.051, 5.283]$. At -10 dB alone the interval is $[21.354, 36.979]$ percentage points. Learned seed-specific rates there are 5.47%, 5.47% and 7.81%, versus 35.16% for OptShrink.

TABLE XXVIII
POST-HOC ERROR-THRESHOLD SENSITIVITY ON THE STRONGER BANK. RATES AVERAGE SEVEN SNR CELLS; POSITIVE DIFFERENCES FAVOR LEARNING.

NMSE threshold	Learned (%)	In-loop OptShrink (%)	OptShrink minus learned 95% interval (points)
0.5	17.00	15.85	$[-1.935, -0.409]$
1	0.89	5.02	$[3.051, 5.283]$
2	0.00	0.00	—

The conclusion depends on the error threshold (Table XXVIII). At threshold 0.5, OptShrink has fewer exceedances; at threshold 2, neither method has any observed exceedance. Zero observed events do not establish zero population risk. These exploratory, pointwise intervals retain paired worlds and fixed models and are not adjusted for multiple thresholds. Learning therefore does not dominate the whole error distribution. The original severe-error definition and primary NMSE metric are unchanged.

The post-review timing run measures all five methods together on the same independent 24-world SNR 5 dB timing bank, with three repetitions. Mean times are 5.16 ms for one adaptive model, averaged across three trained models, 5.68 ms for in-loop OptShrink, 3.87 ms for final-only OptShrink, and 119.08/358.07 ms for one-/four-sweep HS-Rician. Initialization and setting selection or lookup are included; model loading and file I/O are excluded. These are new paired timing measurements, not averages with the earlier timing run. Backend differences remain. Learning avoids the hard method's per-case search, but has no large speed advantage over offline-tuned OptShrink.

A. What the Uncertainty Includes

The new audit uses 4,000 bootstrap draws, resampling 128 whole worlds within each cell. A draw keeps all paired methods and learned seed outputs together. NMSE first sums errors and channel energies within each cell, converts their ratio to dB, averages fixed-model scores, and then equally averages the seven cells. Failure rates instead count per-world error/energy ratios above the stated threshold and average fixed-model rates. Training seeds are not independent test observations. The intervals are pointwise, conditional on the saved models and exploratory. Thresholds 0.5 and 2 were added for sensitivity analysis; the original threshold 1 was not selected to maximize a new contrast. No risk dominance or safety-critical guarantee is inferred. A degenerate empirical bootstrap when there are no events is not evidence of known zero event probability; such intervals are omitted from the table.

B. Verification and Numerical Scope

The run checks original source and checkpoint hashes, saved observations and truth hashes, and all 1,152 trial identities. It recomputes the old prediction errors from channel arrays, rather than copying headline statistics. Four-sweep replay agrees exactly with the existing solver; full-rank bypass is exactly equal for one and four sweeps. Each added prediction retains its channel, selected rank, all candidate scores, trial ID and processing time. A run lock prevents duplicate writers, and completed cases are reused on resume. Earlier result files are not overwritten. The accompanying evidence archive contains this follow-up's protocol, code, source hashes, predictions and aggregate outputs. The large original observation/checkpoint bank is retained in the separate Paper 2 reproduction archive. These checks validate the recorded computation, not external replication or hardware performance.

The rest of this supplement retains the earlier learned-study evidence. Statements about its original timing samples, baselines and data counts refer to those separately labelled protocols, not the new follow-up.

XVI. HISTORICAL: LOCAL RECOVERY AND REFERENCE CALIBRATION

For one array element, let $g, s_p \in \mathbb{C}^K$, with $\mu_p = g^T s_p + b_p$ and noiseless amplitude $a_p = |\mu_p|$. Use real coordinates $x = [\operatorname{Re} g^T, \operatorname{Im} g^T]^T$. When every $\mu_p \neq 0$, differentiation gives

$$da_p = \operatorname{Re}\{\mu_p^* s_p^T dg/|\mu_p|\}, \tag{14}$$

$$A_p = [\operatorname{Re} c_p^T, -\operatorname{Im} c_p^T], \qquad c_p = \mu_p^* s_p/|\mu_p|. \tag{15}$$

Here $A \in \mathbb{R}^{P\times 2K}$ is the real amplitude Jacobian.

Local statement. If $\operatorname{rank} A = 2K$, the noiseless amplitude map is one-to-one in a sufficiently small neighborhood of this g. To see this, choose $2K$ independent rows. Their square Jacobian is invertible, so the inverse-function theorem supplies a local inverse for that measurement subset. The remaining measurements cannot remove this local injectivity. This is not a global uniqueness theorem: distant channels may give the same measurements. A rank-deficient Jacobian also does not, by itself, rule out all forms of nonlinear local injectivity.

For small perturbations at a full-rank point, least-squares inversion has first-order sensitivity bounded by $1/\sigma_{\min}(A)$. Thus $P \geq 2K$ is necessary for regular full-rank local recovery of an unrestricted row. It is not sufficient. Full row rank of S permits the EM linear solve but does not ensure full rank of A. These are standard local arguments in an affine phase-retrieval setting, not a new identifiability theorem [4]. Sparse channel constraints require their own parameterization and Jacobian; a path-count condition is not derived here.

A. Rician Information

For known complex variance $v > 0$ and $a > 0$, the amplitude score is

$$h(z, a, v) = -2a/v + (2z/v)I_1(2za/v)/I_0(2za/v). \tag{16}$$

Define $w(a, v) = \mathbb{E}[h(Z, a, v)^2]$, with expectation under the Rician observation law. For independent pilots the real Fisher matrix is

$$J = A^T \operatorname{diag}\{w(a_p, v)\}_{p=1}^P A. \tag{17}$$

Each magnitude observation contributes an outer product of rank at most one, not a full-rank complex information block. For finite $v > 0$ and nonzero amplitudes these weights are positive. At a zero field the regular amplitude-coordinate

derivation is not used; first-order information about the Cartesian mean vanishes. When the assumptions hold, the rank and conditioning of J give a local diagnostic. Neither a small bound nor a well-conditioned matrix at the true channel guarantees that an iterative estimator reaches the correct solution.

B. What a Reference Does and Does Not Fix

With $B = 0$, g and $e^{j\phi}g$ have identical amplitudes for every ϕ. The Jacobian has the phase direction jg in its null space for nonzero g. A nonzero known reference can remove that symmetry, but arbitrary reference and pilot patterns still require a rank check. For example, if $b_p = c^T s_p$ for some fixed c, then $g' = e^{j\phi}(g+c)-c$ produces the same amplitudes. This can occur with a full-row-rank pilot matrix and a nonzero reference. RSR 0 dB in our experiments means a nonzero reference equal in expected power to one user; it is not the absent-reference case.

There is a separate calibration ambiguity. If the estimator assumes $B' = e^{j\phi}B$, then $G' = e^{j\phi}G$ gives

$$|G'S + B'| = |e^{j\phi}(GS + B)| = |GS + B|. \tag{18}$$

The likelihood therefore cannot identify a constant reference-phase error from these observations alone. With error measured in the original reference frame, even this otherwise exact rotated estimate has NMSE $|e^{j\phi} - 1|^2 = 4\sin^2(\phi/2)$. At 10° this is about 0.0304 (-15.2 dB). This is an illustrative exact-fit ambiguity, not a lower bound on every biased estimator. A smaller withheld loss cannot certify correct absolute phase or recover an unknown reference phase. Time-varying phase errors generally cannot be absorbed into one constant channel rotation. A separate static calibration diagnostic estimates a shared positive amplitude multiplier and noise variance under stated assumptions. It cannot remove this unknown-phase ambiguity.

XVII. Historical: Filter Properties and Computation

Let $H = U\,\mathrm{diag}(\sigma_i)V^H \in \mathbb{C}^{m\times q}$, $m \geq q$, and

$$F_\tau(H) = H - \tau^2 H(H^H H + \tau^2 I)^{-1}, \qquad \tau > 0. \tag{19}$$

Its singular values are $f_\tau(\sigma) = \sigma^3/(\sigma^2 + \tau^2)$. Thus $0 \leq f_\tau(\sigma) \leq \sigma$ and

$$\|F_\tau(H)\|_F \leq \|H\|_F, \qquad \|F_\tau(H) - H\|_F \leq \sqrt{q}\,\tau/2. \tag{20}$$

The second inequality follows because $\sigma - f_\tau(\sigma) = \sigma\tau^2/(\sigma^2+\tau^2) \leq \tau/2$.

For a channel vector, let d_i count its occurrences in the Hankel matrix and define $\|g\|_D^2 = \sum_i d_i|g_i|^2 = \|\mathcal{H}(g)\|_F^2$. Antidiagonal averaging followed by lifting is the orthogonal projection onto Hankel matrices in Frobenius norm. Therefore for fixed realized $\eta \in [0,1]$ and $\tau > 0$,

$$g^+ = (1-\eta)g + \eta\mathcal{H}^\dagger F_\tau(\mathcal{H}(g)), \tag{21}$$

$$\|g^+\|_D \leq \|g\|_D, \qquad \|g^+ - g\|_D \leq \eta\sqrt{q}\,\tau/2. \tag{22}$$

This bounds the size of one correction, not its error relative to the unknown true channel. It does not prove a contraction between two different inputs, particularly when the controller changes η and τ with the input. It also does not prove convergence of the full estimator. For fixed H, $\tau \to 0$ gives $F_\tau(H) \to H$ and $\tau \to \infty$ gives zero. No low-noise limit is asserted because the learned threshold need not vanish with noise.

The matrix-solve shrinker is smooth for $\tau > 0$, including repeated singular values. However, the controller also uses ordered largest and tail singular values. These features need not be differentiable at every singular-value tie. Finite-gradient numerical checks do not establish global differentiability of the whole network.

A. Why the Rician Measurement Update Is an EM Step

Let the unobserved complex field be $Y = GS + B + W$ and write each entry as $Y_{np} = Z_{np}e^{j\Phi_{np}}$. The receiver observes Z, not the phase Φ. For a current predicted field M, the conditional phase density is

$$p(\phi \mid z, M) = \frac{\exp\{\kappa\cos(\phi - \arg M)\}}{2\pi I_0(\kappa)}, \qquad \kappa = 2z|M|/v. \tag{23}$$

This is a von Mises distribution. Its first circular moment gives

$$Q = \mathbb{E}[Y \mid Z, M] = Z\frac{I_1(\kappa)}{I_0(\kappa)}e^{j\arg M}. \tag{24}$$

At $M = 0$, the phase is uniform and $Q = 0$. These are conditional complex fields, not estimates of the mean magnitude. Since $|Y|^2 = Z^2$ when Z is fixed, expanding $\mathbb{E}[\|Y - GS - B\|_F^2 \mid Z, M]$ leaves $\|GS + B - Q\|_F^2$ plus terms independent of the candidate G. Least squares therefore gives $(Q-B)S^H(SS^H)^{-1}$. This derivation uses independent proper complex Gaussian noise and a supplied positive variance; it does not use a strong-reference or high-SNR approximation.

B. Relaxed EM Is Not the Same as the Full Stage

For fixed posterior field Q and supplied variance $v > 0$, the negative EM surrogate, up to constants, is $\Psi(G) = \|GS + B - Q\|_F^2/v$. Its minimizer is G_{EM} when S has full row rank. The normal equations give

$$\Psi(G) - \Psi(G_{\mathrm{EM}}) = \|(G - G_{\mathrm{EM}})S\|_F^2/v. \tag{25}$$

For $G_\gamma = G + \gamma(G_{\mathrm{EM}} - G)$, the surrogate decrease is

$$\Psi(G) - \Psi(G_\gamma) = \gamma(2-\gamma)\|(G - G_{\mathrm{EM}})S\|_F^2/v \geq 0 \tag{26}$$

for $0 \leq \gamma \leq 2$. Thus even an over-relaxed step within this interval is a generalized EM step for this quadratic M-step, in exact arithmetic. The supplied-model likelihood is nondecreasing. This result does not extend automatically to the subsequent Hankel correction, nor does it guarantee channel-error reduction, global recovery or convergence of the complete learned estimator. The secondary GD core is not covered by this particular argument.

TABLE XXIX
LEADING WORK PER STAGE, EXCLUDING INITIALIZATION

Operation	Leading order
Predicted fields and EM products	$O(NKP)$
EM solve, current implementation	$O(NK^2 + K^3)$
Spectral controller features	$O(Kmq^2)$
Smooth matrix filter	$O(Kmq^2 + Kq^3)$
Antidiagonal averaging	$O(Kmq)$
Small controller	$O(K \cdot 178)$

C. Operation Counts

For $S \in \mathbb{C}^{K\times P}$ and $G \in \mathbb{C}^{N\times K}$, forming SS^H costs $O(K^2P)$. Spectral initialization forms N weighted $(K+1)$-dimensional Gram matrices and eigendecompositions, costing $O(NP(K+1)^2+N(K+1)^3)$. These are per observation case, not per stage. The inverse-trace noise proxy adds $O(K^3)$ once.

Here $q = \lceil N/2 \rceil$, $m = N - q + 1$. Caching the pilot-Gram factorization could remove repeated K^3 factorizations, but the quoted timings use the existing implementation. The filter avoids differentiating hard rank truncation; it does not remove the SVD needed for controller features. At large N, both dense spectral features and the dense solve scale cubically. A small parameter count is not a small operation count.

For HS-GS, with $R = q$ candidate ranks, $T_s = 25$ selection updates, $T_f = 100$ refit updates and $C = 4$ sweeps, the dominant repeated work is of order

$$(RT_s + T_f)\,[NKP + NK^2 + K^3 + CKmq^2]. \qquad (27)$$

Selection uses fewer pilot columns and has additional scoring and initialization work. This expression is a work estimate, not an exact FLOP count or a prediction of the measured speedup. Bessel functions, library choices, memory traffic and batching also matter. At $N = 32$, HS-GS uses 400 selection and 100 refit updates, compared with ten learned stages. Accuracy at those different budgets, not equal iteration count, is the reported comparison.

XVIII. HISTORICAL: EXISTING DIMENSION TESTS AND CONTROLLER INPUTS

The original study already contains one-factor-at-a-time tests, with 400 cases per condition and weights fixed. Except for the changed quantity, the reference condition is $N = 32, K = 3, P = 30$, SNR 5 dB and RSR 12 dB. The pilot sweep's smallest value is ten for three users, not six. It does not test the boundary $P = 2K$. The alternative channels use four clusters of ten rays within $\pm 5^\circ$, ten equal-power paths, or ten paths with geometrically decreasing powers. They match base-channel energy and are tested only at SNR 5 dB and RSR 12 dB. These alternatives are simulation populations, not measured scattering distributions.

The full controller receives a spectrum summary, residual/noise proxies and size/stage information. A large spectral tail can reflect either noise or real paths; residuals can reflect noise, a wrong channel, or a wrong receiver model. The features provide context but no unambiguous failure detector. No sign constraint makes the threshold monotone in SNR, RSR or path count. The threshold and mixing can compensate for each other, so we inspect the actual relative channel correction as well as both outputs.

TABLE XXX
ALL ORIGINAL MAIN-STUDY COMPARATORS, UNCHANGED

Method	NMSE (dB)	Time (ms)	Severe (%)
EM adaptive (primary)	-14.42	8.24	0.45
EM learned steps only	-10.41	2.98	22.11
EM global filtering	-12.43	5.61	19.19
EM constant-output control	-9.19	5.61	11.42
GD adaptive (secondary)	-14.59	8.00	0.85
GD learned steps only	-10.44	2.73	21.15
GD global filtering	-12.57	5.38	18.37
GD constant-output control	-9.95	5.36	12.36
HS-GS, rank search	-12.74	525.62	14.25
EM, 10 updates	-10.23	4.10	20.64
EM, 100 updates	-10.28	17.01	23.57
Rician GD, 10 updates	-9.31	4.17	17.86
Rician GD, 100 updates	-10.29	17.40	23.50
Rician GD, stopping rule	-10.27	45.37	23.75
Amplitude GD, 10 updates	-8.80	3.29	20.21
Amplitude GD, 100 updates	-9.65	8.87	26.96
Amplitude GD, stopping rule	-9.61	20.93	27.11
Rician GD, hard Hankel	-11.06	47.39	20.93
Linearized least squares	-6.92	1.06	19.86

Accuracy averages seven primary SNR cells. Time is the original batch-one median at SNR 5 dB, RSR 12 dB. These are not the new OptShrink comparison's data or timing sample.

TABLE XXXI
ORIGINAL TRANSFER RESULTS: GAIN OVER HS-GS IN DB

Changed quantity	Gain	Pointwise 95% interval
$N = 8$	-0.106	$[-0.286, 0.077]$
$N = 16$	1.223	$[1.086, 1.361]$
$N = 64$	0.165	$[0.024, 0.307]$
$K = 2$	0.078	$[-0.073, 0.228]$
$K = 4$	1.529	$[1.415, 1.639]$
$P = 10$	2.334	$[2.136, 2.531]$
$P = 20$	1.427	$[1.273, 1.577]$
Noise input -3 dB	1.226	$[1.095, 1.359]$
Noise input $+3$ dB	1.050	$[0.901, 1.193]$

A. Original Follow-up Comparisons

The earlier follow-up trains 15 models and evaluates two fresh sets, each with seven SNR settings and two difficult settings of 200 cases. The six primary comparisons adjust for multiple testing within each set. The two new-data adaptive runs use independently generated training data, one fixed training seed per dataset. They are distinct from the three main seeds sharing one dataset and from the new reviewer diagnostics.

The supervised recipe holds architecture, input pilots and training budget fixed but trains on channel targets with mean linear NMSE. Selecting its checkpoint by withheld likelihood reduces the gap to approximately 0.73 dB; pooling linear errors across SNRs reduces it to about 0.36 dB. It wins in some cases, including two paths. Its transfer from 21 to 30

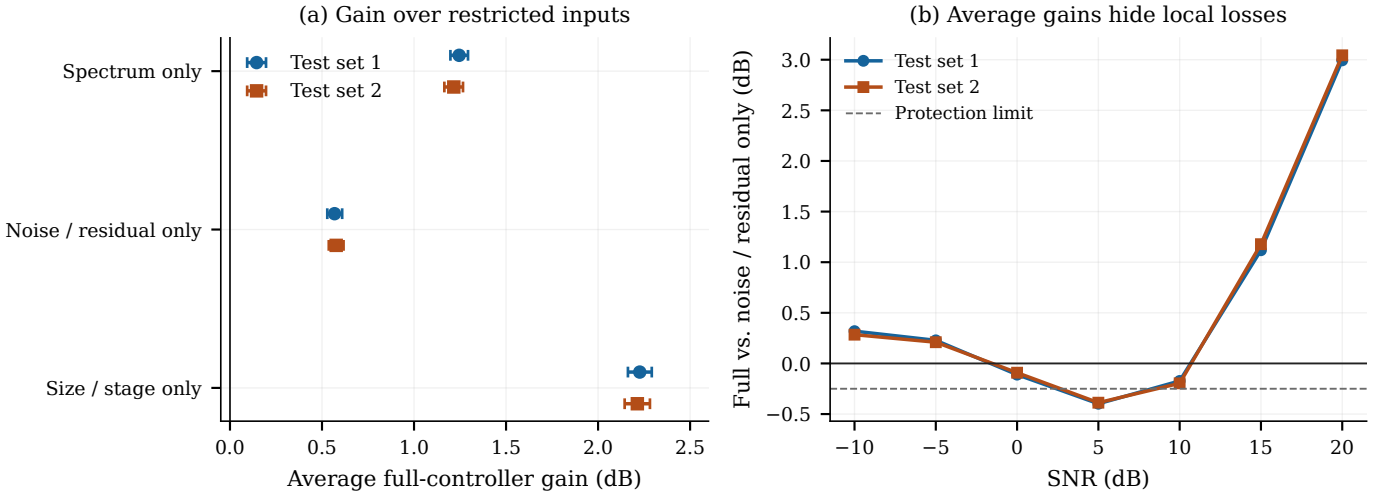


Fig. 8. Original smaller-budget input ablation, unchanged. Full inputs help the average but fail the declared 0.25 dB protection condition. These are separate models trained with 4,096 cases, 12 epochs and three seeds; they are not replacements for the main 16,384-case models.

TABLE XXXII
ORIGINAL FOLLOW-UP AVERAGE GAINS IN DB, UNCHANGED

Comparison	Set 1	Set 2
New-data adaptive vs. global	2.252	2.185
New-data adaptive vs. no filtering	4.250	4.191
New-data adaptive vs. HS-GS	1.939	1.837
Original adaptive vs. supervised	1.015	1.012
Original adaptive vs. final-only	0.697	0.694
Fixed-open mix vs. original adaptive	0.064	0.066

TABLE XXXIII
EXPLORATORY SEVEN-SNR AVERAGES, 32 FRESH CASES PER CELL

Method	NMSE (dB)
EM100, no final filter	-10.162
EM100 + final hard rank six	-11.811
EM100 + final cubic filter	-11.648
EM100 + final OptShrink rank seven	-13.691
HS-GS, residual rank score	-12.778
HS-GS, Rician rank score	-12.961
Frozen learned model, stage-five prefix	-13.353
Frozen learned model, all ten stages	-14.243

pilots was less consistent in validation replay. This comparison does not establish a general benefit of withholding labels. The final-only comparison retrains that alternative and its separate timing sample gives 4.65 versus 2.05 ms; those values do not replace the main timing table. Fixing mixing at one retains all 188 stored parameters and worsens two of three seeds on two-path cases, despite its small average gain.

XIX. HISTORICAL: ADDITIONAL REVIEWER DIAGNOSTICS

These checks were specified after the main study and use frozen selected EM models (seeds 9101–9103). No learned weights or original results were changed. Seven SNR cells and two stress cells each contain 32 fresh worlds: 288 distinct worlds. Eight input-mismatch conditions reuse the same 32 SNR 5 dB worlds, giving 544 condition-cases, not 544 independent worlds. Classical and learned methods share the channel, pilots and noise. All tests use $N = 32, K = 3, P = 30$; the main RSR is 12 dB. The extra cells have RSR 0 dB or two paths, at SNR 5 dB.

The small sample, post-review design and fixed checkpoints make these exploratory checks. The intervals resample whole cases with common weights across methods and learned seeds, separately within each cell. They are unadjusted, conditional intervals, not a new confirmatory success test. The original 400-case cells and their intervals remain unchanged.

A. Same Candidates, Different Rank-Selection Loss

HS-GS fits ranks 1–16 on the first 21 pilots using 25 EM updates and four Cadzow sweeps. Exactly the same candidate channels are scored on the last nine pilots using either squared magnitude residuals or Rician negative log-likelihood. Each selected rank is refitted with 100 updates and all 30 pilots. This changes only the selection score, not the fit budget or observation count. A replay against the original residual-ranked solver agreed exactly on a fresh case; an independent likelihood check agrees with SciPy's Rician density up to observation-only constants.

Across the seven SNR cells, Rician scoring improves HS-GS by 0.182 dB, with interval $[0.104, 0.267]$ dB. The frozen learned estimator retains a gain of 1.282 dB, with interval $[1.092, 1.485]$ dB. This score difference explains part, but not all, of the gap in this small check.

B. A Stronger Final-Filter Control

A separate bank of 72 development worlds (eight per SNR/stress cell) selects one global setting for each final-filter method. Selection uses only the withheld Rician likelihood after EM100 on 21 pilots. Hard truncation and OptShrink search ranks 1–15; cubic filtering searches relative thresholds $\{0.01, 0.03, 0.1, 0.3, 1, 3\}$. Every method also has a no-correction option. Candidate counts differ and the grids are

TABLE XXXIV
INPUT MISMATCH AT SNR 5 DB: LOWER NMSE IS BETTER

Assumed input error	Learned (dB)	HS-Rician (dB)	Learned loss vs. matched (dB)
None	-14.043	-12.496	0.000
B amplitude -10%	-11.613	-9.799	2.430
B amplitude $+10\%$	-12.252	-11.328	1.791
B phase -10°	-11.685	-10.730	2.358
B phase $+10^\circ$	-11.572	-10.564	2.471
Phase ramp $0 \to -10^\circ$	-13.267	-12.029	0.776
Phase ramp $0 \to +10^\circ$	-13.204	-11.774	0.839
Noise variance -6 dB	-13.806	-12.567	0.238
Noise variance $+6$ dB	-10.995	-12.404	3.048

small; this is not an exhaustive equal-search-budget comparison. The selected settings are rank six, relative threshold 0.3 and OptShrink rank seven, respectively. The settings are frozen before reading test observations, then used after EM100 with all 30 pilots. None use true channels for tuning.

OptShrink uses the empirical spectral-tail transform in the original algorithm [1], including rectangular zero directions and reduced tail dimensions. A rank equal to full capacity has no tail and is excluded; no correction is a separate candidate. At a tied boundary within a relative squared gap of 10^{-13}, the affected weight is set to its zero-weight limit. Primitive checks cover zero tail, scaling, transpose symmetry, direct rectangular trace evaluation, zero input and the tie rule. These are numerical safeguards, not a new optimality result.

In the matched one-pass test, OptShrink beats the cubic filter by 2.042 dB on the seven-SNR average. Cubic shrinkage is therefore not generally the best structural operation. The complete frozen learned estimator has a 0.552 dB average gain over EM100 plus OptShrink, with conditional interval $[0.432, 0.673]$ dB. It loses 1.040 dB at SNR 20 dB and 2.408 dB at RSR 0 dB, but gains 1.830 dB on two-path cases against this globally selected rank-seven control. That last result does not remove the two-path loss against per-case rank-selected HS-GS.

These comparisons have different update budgets, training/development budgets and filtering schedules. They do not identify the best possible OptShrink-based estimator or reproduce iterative LRHD. No controlled runtime comparison was added; the original 64-fold timing ratio applies only to the original HS-GS implementation.

C. Reference and Noise Inputs

The physical observations stay fixed. For amplitude errors the estimator receives $B_{\text{in}} = aB_{\text{true}}$ with $a = 0.9$ or 1.1; these are amplitude, not power, multipliers. Constant phase errors use $B_{\text{in}} = e^{j\delta}B_{\text{true}}$ with $\delta = \pm 10^\circ$. The ramp uses $B_{\text{in},np} = B_{\text{true},np}e^{j\delta p/(P-1)}$, $p = 0, \ldots, P-1$: the assumed phase error grows from zero to $\pm 10^\circ$. Noise errors supply $v_{\text{in}} = v_{\text{true}}10^{\Delta/10}$, $\Delta = \pm 6$ dB. These are input-mismatch stress tests, not measured hardware drift. They test errors relative to a fixed true reference, not a dynamically changing atomic receiver or a joint calibration algorithm.

Supplying noise variance 6 dB too high worsens the learned estimate by 3.048 dB relative to matched inputs, and makes it 1.409 dB worse than Rician-ranked HS-GS on these paired cases. This is a new observed failure, not evidence of general variance robustness. Amplitude errors also cause clear losses. Constant reference-phase errors rotate every estimated channel by the same phase; replay verifies the equivariance and unchanged phase-aligned errors. Alignment uses truth only as a diagnostic and is not an available inference correction. Absolute-frame errors in the table remain the reported errors.

D. Depth and Controller Trajectories

The stage-five output is taken from the unchanged ten-stage forward pass, with its original stage feature $(t+1)/10$. It is not a five-stage model retrained with $(t+1)/5$. Completing ten stages improves the seven-SNR average by 0.890 dB, with interval $[0.815, 0.964]$ dB; it does not help every cell. This prefix experiment does not determine an optimal depth or answer whether a trained twenty-stage model is better.

At the ordinary SNR 5 dB condition the mean relative threshold falls from about 0.915 at stage one to 0.362 at stage ten. Mean mixing is about 0.99996. Thus the useful adaptation is primarily in threshold strength, not abstention. Large relative corrections in a failure condition do not prove which feature or prior caused the failure.

XX. HISTORICAL: CONTROLLED IN-LOOP OPTSHRINK COMPARISON

This experiment uses the frozen original three EM models. A separate development set has 32 worlds at each of seven SNRs. For both in-loop and final-only OptShrink, the candidate grid includes ranks 1–15, mixing 0.25, 0.5 or 1, update budgets 10, 20, 50 or 100, and no filtering. Candidates fit 21 pilots; the other nine score Rician likelihood. Noise-dependent settings are frozen before generating 128 fresh worlds at each SNR and at each of two stress settings, giving 1,152 test worlds. All test methods use all 30 pilots. Reference strength is 12 dB and channels have three to seven paths, except in the labelled stress cells.

Unlike the first final-filter diagnostic, this control shrinks the lifted channel after each EM update and averages its antidiagonals. It is an OptShrink-based Rician reconstruction, not full LRHD on directly observed noisy Hankel matrices. Hankel lifting and nonlinear reconstruction produce dependent errors, so matrix-denoising optimality guarantees are not claimed. Both noise-dependent and global settings are retained in the results.

The retained narrow-population figure includes these saved results and the added one-sweep comparison. The per-SNR intervals are conditional paired-case intervals, with no per-SNR multiplicity adjustment.

The selected in-loop control uses ten iterations up to SNR 10 dB and twenty at 15 and 20 dB. The learned model loses 0.143 dB on the declared seven-SNR average. It gains 0.723 dB at -10 dB but loses 0.796 dB at 20 dB. Its severe-error rate is 0.89%, versus 5.02% for the selected in-loop method. At -10 dB these rates are 6.25% and 35.16%; the

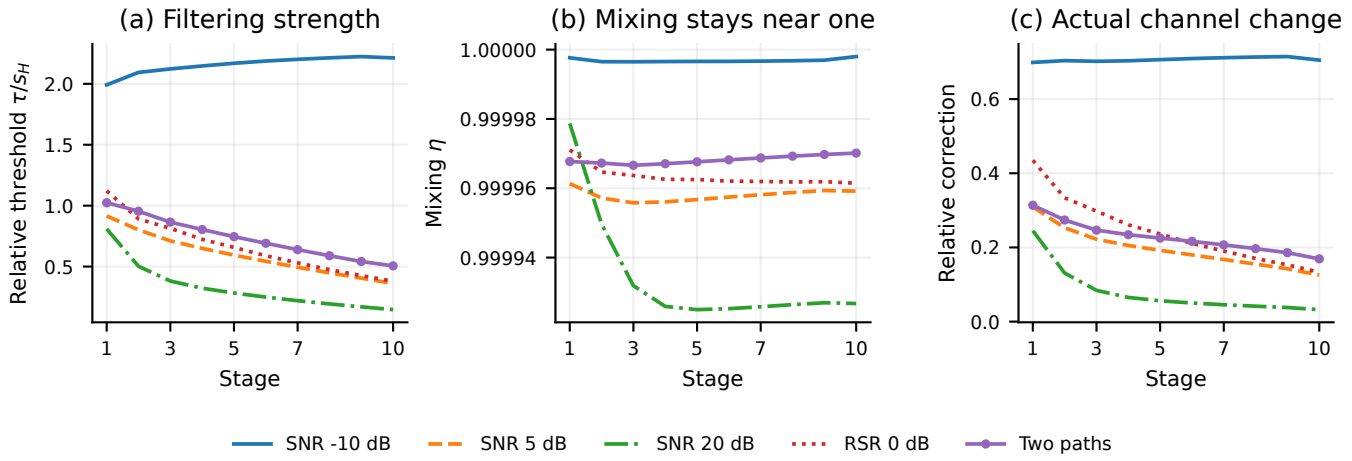


Fig. 9. New frozen-model trajectories, averaged over 32 cases, three model seeds and users. Main SNR curves use RSR 12 dB; weak-reference and two-path curves use SNR 5 dB. All other settings are unchanged. Mixing stays near one, while threshold and actual correction vary. Curves describe outputs; they do not establish causal feature importance or a failure detector.

other primary SNR cells have none in this sample. The lower mean per-case linear error of the learned method is a different metric, not a reason to replace the original primary metric after seeing the results.

At RSR 0 dB and SNR 5 dB, the selected in-loop control has NMSE -5.026 dB, versus -7.069 dB for learned and -8.307 dB for HS-Rician. However, its 50- and 100-iteration controls reach -8.790 and -8.981 dB. The apparent learned advantage over the selected control therefore does not establish superiority over the in-loop family at weak references. For two paths, HS-Rician reaches -17.833 dB, learned -16.814 dB and selected in-loop -16.407 dB. No algorithm wins every comparison.

End-to-end single-thread CPU timing at SNR 5 dB includes initialization, noise-level lookup and the selected reconstruction. Mean times are 5.67 ms for one learned model, averaged across three trained models, and 6.20 ms for selected in-loop OptShrink. Final-only filtering takes 3.99 ms and HS-Rician 356.65 ms in this timing sample. These are not ensemble inference times, embedded-device measurements or GPU benchmarks. The original 64-fold ratio applies only to the earlier HS-GS comparison. Saved outputs were recalculated; independent OptShrink primitive replays agree within 5×10^{-15} and selected learned replays agree exactly.

XXI. Historical: Static Noise and Reference Estimation

This exploratory check estimates a single positive reference-amplitude multiplier a shared by the array and a shared complex noise variance v. Reference shape and phase remain supplied. It does not estimate per-antenna drift or choose new measurements. With posterior field Q, let $R = I - S^H(SS^H)^{-1}S$. One conditional least-squares update is

$$a^+ = \mathrm{clip}_{[0.25,4]} \frac{\mathrm{Re}\langle BR, QR\rangle_F}{\|BR\|_F^2}, \tag{28}$$

$$G^+ = (Q - a^+B)S^H(SS^H)^{-1}, \tag{29}$$

$$v^+ = \mathrm{clip}_{[10^{-6},10^3]}\, \mathrm{mean}\{Z^2 + |M^+|^2 - 2\,\mathrm{Re}[(M^+)^*Q]\}, \tag{30}$$

where $M^+ = G^+S + a^+B$. Bounds use the normalized simulation units and are algorithm assumptions, not measured receiver limits. If $BR = 0$, the gain is confounded with the channel and the joint fit is rejected. This denominator check is not a global identifiability proof for the full nonlinear problem.

Three choices—unchanged inputs, noise estimation, and joint gain/noise estimation—are fitted on 21 pilots for 5, 20 or 100 iterations. The other nine choose a setting, which is refitted on all 30 observations. The selection loss includes $\log v$:

$$\ell_{\mathrm{nuisance}} = (z - |\mu|)^2/v - \log I_{0e}(2z|\mu|/v) + \log v. \tag{31}$$

Only $-\log(2z)$ may be omitted when comparing different variances. All nine candidates and the final refit count toward processing cost. The calibration preprocessing averages 45.05 ms per case in this CPU check, before the unchanged learned estimator runs. It is not a free robustness improvement. An EM control with the same held-out stopping choice separates stopping effects from nuisance estimation. Corrected inputs are also passed to the unchanged learned model; there is no retraining on these cases.

There are 32 worlds per SNR/reference cell, at SNR $-5, 5, 15$ dB and RSR 0 or 12 dB: 192 independent worlds. Seven assumed-input conditions reuse them, giving 1,344 condition-cases, not independent worlds. The conditions are correct inputs, variance errors of ± 6 dB, reference-amplitude errors of $\pm 10\%$, joint $+10\%/+6$ dB error, and constant phase error $+10^\circ$.

Average improvement does not make the correction safe in every cell. With weak reference at SNR -5 dB and a $+6$ dB variance error, it worsens learned NMSE by 1.048 dB. At SNR

TABLE XXXV
STATIC CALIBRATION: GAIN OF CORRECTED OVER SUPPLIED INPUTS

Input condition	Mean gain (dB)	Worst cell gain (dB)
Correct inputs	-0.013	-0.054
Variance -6 dB	0.832	0.138
Variance $+6$ dB	0.997	-1.048
Amplitude -10%	1.863	-0.072
Amplitude $+10\%$	1.449	-0.413
Joint $+10\%/+6$ dB	2.872	-0.746
Phase $+10^\circ$	-0.008	-0.048

Equal average over six SNR/RSR cells. Positive gain means lower error. These are descriptive point estimates, not a simultaneous robustness guarantee.

15 dB with strong reference, a $+10\%$ amplitude error is largely corrected: NMSE changes from -16.346 to -24.107 dB; the true-input diagnostic is -24.253 dB. These are simulation input errors, not observed hardware behavior. Unknown constant phase remains unresolved because channel and reference can rotate together. Likelihood improvement cannot certify absolute phase or reduced channel error. The original known-calibration estimator remains the primary method; this check is not a universal calibration remedy.

XXII. HISTORICAL: TIGHT PILOT COUNTS AND WEAK REFERENCES

The new pilot sweep has $P = 6, 8, 10, 12, 20, 30$, $K = 3$, SNR $-5, 5, 15$ dB, RSR 12 dB and 32 fresh worlds per cell, or 576 worlds. Learned weights and OptShrink settings are transferred from $P = 30$, without tuning on these tests. HS-Rician fits ranks on $\mathrm{round}(0.7P)$ pilots, scores the remainder, then refits all P. Its rank-fitting subset can be below $2K$ even when P reaches it. Structural coupling across array elements means the unrestricted single-row count is not, by itself, an identifiability theorem for the structurally constrained model.

At SNR 5 dB, learned NMSE is -2.197 dB with six pilots and -14.332 dB with thirty. At SNR -5 dB and six pilots it is 4.089 dB, showing substantial failure. All sampled nonzero-reference full-data row Jacobians have rank six. Nevertheless, at SNR 5 dB the median smallest singular value is 0.172 with six pilots and 2.488 with thirty. Local full rank is not enough for stable finite-noise estimation or global recovery. The cells use different independent worlds; the method comparisons within each cell are paired.

A second check shares the same channel, pilots and complex noise across absent reference and RSR $-10, 0, 5, 12$ dB, at SNR 5, 20 and 60 dB. There are 32 independent worlds per SNR and 480 condition-cases. The SNR 5 dB reference check shares a generation seed and channels with one tight-pilot cell. The two diagnostic phases are not independent replications of each other and are not pooled for uncertainty estimates. At zero reference, sampled Jacobians have rank five and absolute phase is ambiguous. At nonzero reference they have rank six in this sample. At SNR 20 dB and RSR -10 dB, ordinary EM gives -1.773 dB NMSE from spectral initialization but -21.036 dB when initialized at truth. The latter is an oracle diagnostic only. The gap supports an initialization or optimization difficulty; it does not prove global identifiability or exclude controller problems.

A. A Nearly Noiseless Failure

SNR 60 dB lies far outside the training range. Even at RSR 12 dB the frozen learned model gives -6.647 dB, while EM100 gives -65.715 dB. A post-hoc intervention on the same 32 cases disables only learned filtering and retains the learned step sizes; its NMSE is -44.020 dB. Starting the full learned model at the true channel still gives -6.653 dB. Thus filtering can spoil an already accurate estimate in this setting; the failure is not explained solely by a bad initializer. Independent NumPy arithmetic for the no-filter intervention agrees with the PyTorch result within 5×10^{-15}. This is an out-of-range failure and a post-hoc diagnostic, not an independently confirmed switching policy. No low-noise consistency or generally safe filtering claim is made.

XXIII. HISTORICAL: SEPARATELY TRAINED DEPTH, WIDTH AND TRAINING CONTROLS

This reduced-budget study answers the depth and capacity questions using new training, not a prefix of the original network. Each dataset contains 4,096 training and 448 independent validation worlds. Training uses 12 epochs, batch size 32 and Adam with learning rate 0.001. There are seven SNRs from -10 to 20 dB, RSR 0, 5, 12 and 20 dB, and either random three-to-seven-path or fixed two-path channels. This gives 56 conditions, with $N = 32, K = 3, P = 30$. Unlike the original training mixture, this one includes RSR 0 dB. Its scores must not be pooled with the original study.

Seven variants share dataset zero and three initialization seeds: the ten-stage, width-16 baseline; depths five and twenty; widths eight and 32; two experts; and curriculum training. The baseline also trains on two additional independent datasets with two initialization seeds each, giving 25 jobs and 300 completed epochs. Initialization seeds do not create independent training populations. Training has no channel targets. Validation selects the checkpoint using the equal-condition average of withheld Rician likelihood, with a fixed 21/9 pilot split. The final epoch is also evaluated, not substituted when its test error is better.

Ordinary variants share example order and pilot masks at each epoch. Depth changes the number of learned steps and uses $(t + 1)/T$ as its stage input. The two-expert model combines two width-16 controller outputs with a learned softmax gate driven by the three spectral features. It has 374 parameters, compared with 364 for width 32. The experts start at relative thresholds 0.05 and 0.2 rather than the plain controller's 0.1. They do not recover distinct physical channel components. Architecture and initialization both change; any gain cannot be attributed to gating alone.

The curriculum presents RSR 12/20 examples twice per epoch for six epochs, then RSR 0/5 examples twice per epoch for six epochs. Total exposure per example matches ordinary training. The repeated visits share that epoch's pilot mask, so the number of distinct masks per example differs. This is not

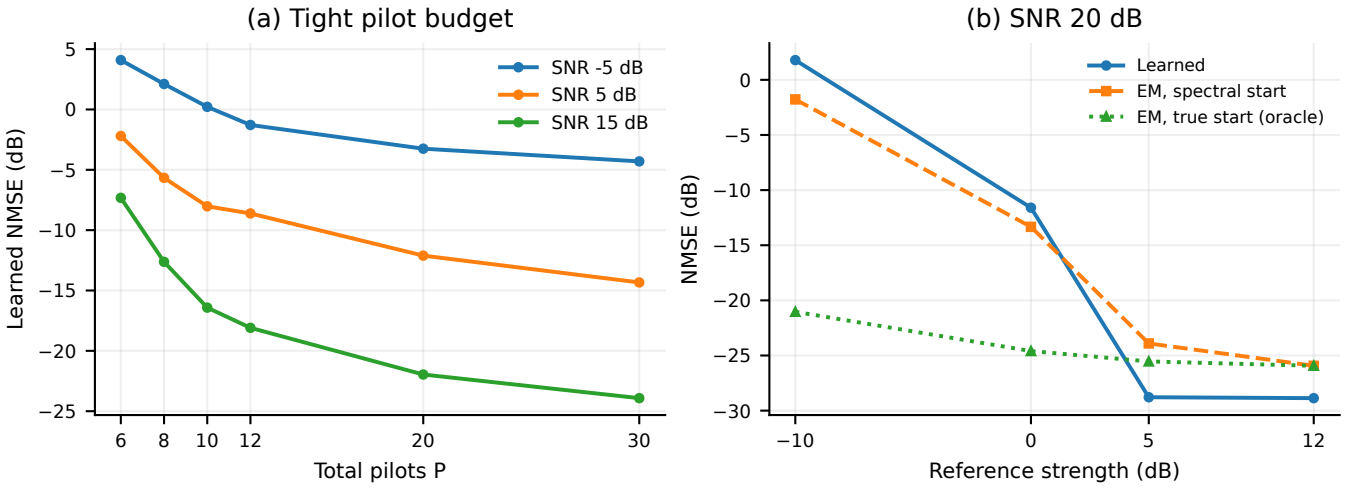


Fig. 10. Exploratory checks, 32 worlds per cell. Left: frozen learned model under changing pilot count. Right: EM initialization diagnostic with the same channel, pilots and noise across reference levels. The true-start curve uses unavailable ground truth and is not a deployable estimator.

an isolated test of ordering alone. Neither channel truth nor test errors determine this schedule.

After all checkpoints were fixed, 64 fresh worlds per condition were generated, giving 3,584 paired test worlds. All 25 selected and 25 final models use the same observations, together with EM100 and the previously frozen in-loop OptShrink rule. That rule transfers from its earlier development mixture; it is not retuned on these new tests. A documented test-seed amendment was made before generation to avoid overlap with diagnostic seeds. Training and selection were unchanged.

A. Accuracy and Cost

The reported macro-average gives each condition and initialization equal weight in dB. Five stages are faster, but lose 1.977 dB against ten. Twenty gain 0.181 dB while increasing mean time from 8.04 to 15.60 ms. Their severe-error rate is also higher: 13.46% versus 10.28% in this 56-condition mixture. The depth comparison is therefore not a simple "more is better" result. Twenty stages improve the RSR 0 dB average by 1.547 dB but worsen one low-SNR cell by 1.586 dB. The absolute severe-error rates use a broader, weaker-reference mixture than the original experiment and are not evidence of a change in its saved results.

Width eight loses 0.731 dB and has almost unchanged processing time: matrix operations, not controller size, dominate this implementation. Width 32 gains 0.065 dB on average and reduces severe errors, but loses 0.970 dB in its worst cell. The two-expert gain is only 0.023 dB overall and 0.179 dB for two-path cells; width 32 gains 0.286 dB on those cells. The experts therefore do not demonstrate a useful advantage over the similarly sized plain controller in this check.

The curriculum gains 0.206 dB overall and 0.255 dB at RSR 0 dB, but loses 2.353 dB at SNR 20 dB, RSR 5 dB and two paths. It does not solve weak-reference or few-path estimation generally. No tested variant improves every cell. The original primary estimator is retained; these test results are not used to choose a replacement.

Timing uses one CPU thread, double precision and batch size one, includes initialization, and repeats 32 shared SNR 5 dB/RSR 12 dB worlds three times in shuffled model order. Means average individual-model inference, not ensemble inference. This timing batch is separate from the original and OptShrink timing batches; cross-batch speed ratios are not claimed. The 25-job training invocation took about 103 minutes on the local CPU, including data preparation and validation, but excluding fresh testing.

B. Variability and Scope

The three independent training datasets give baseline two-initialization macro-averages of -12.450, -12.124 and -12.229 dB. Their individual initialization scores are $(-12.391, -12.509)$, $(-11.824, -12.425)$ and $(-12.143, -12.315)$ dB. These few replicates expose variability but do not estimate broad population uncertainty reliably. All use the same test worlds; repeated predictions are not independent channel realizations.

The paired intervals resample 64 whole cases within each condition 2,000 times, retaining paired methods and averaging the three fixed-model scores. They are exploratory, conditional and unadjusted across six comparisons. Small intervals do not remove uncertainty from training or prove an architecture optimum. Most selected checkpoints are the final epoch; 12 epochs do not establish convergence.

The transferred OptShrink rule gives -12.150 dB and 12.11% severe errors; EM100 gives -8.784 dB and 25.53%. The new baseline's -12.390 dB score cannot reverse the earlier controlled OptShrink result: both the training mixture and test weighting differ, and the classical settings were not developed for this mixture. These are transfer controls, not a new claim of tuned-classical superiority.

C. Saved-Result Audit

All 25 histories, 300 epochs, 50 checkpoint hashes and 2,912 prediction files were checked. The latter comprise 1,400

TABLE XXXVI
SEPARATELY TRAINED CONTROLS: ALL 56 CONDITIONS, EQUAL-WEIGHT DB AVERAGE

Variant	Parameters	Selected NMSE (dB)	Final NMSE (dB)	Gain (dB)	Worst-cell gain (dB)	Severe (%)	Time (ms)
Ten stages / width 16	188	-12.390	-12.297	0.000	0.000	10.28	8.04
Five stages	183	-10.413	-10.413	-1.977	-9.540	6.48	4.25
Twenty stages	198	-12.571	-12.571	0.181	-1.586	13.46	15.60
Width 8	100	-11.658	-11.658	-0.731	-3.070	14.14	8.02
Width 32	364	-12.454	-12.454	0.065	-0.970	4.91	8.04
Two experts	374	-12.413	-12.385	0.023	-0.931	8.31	8.55
Curriculum	188	-12.596	-12.634	0.206	-2.353	6.84	8.03

All rows use dataset zero, three initialization seeds, 4,096 training cases and 12 epochs. Gain is relative to ten stages/width 16; positive favors the variant. Severe rates and gains use selected checkpoints. Time is individual-model mean inference time, not an ensemble time. This protocol differs from the original study.

TABLE XXXVII
INITIALIZATION VARIABILITY AND CONDITIONAL GAINS IN THE REDUCED-BUDGET STUDY

Variant	Selected seed scores (dB)	Selected epochs	Gain interval (dB)	RSR 0 gain	Two-path gain
Ten stages / width 16	-12.391, -12.509, -12.269	12, 12, 11	—	0.000	0.000
Five stages	-10.779, -10.906, -9.554	12, 12, 12	$[-2.137, -1.849]$	-1.500	-1.943
Twenty stages	-12.382, -12.579, -12.751	12, 12, 12	$[0.062, 0.401]$	1.547	0.075
Width 8	-11.906, -11.288, -11.781	12, 12, 12	$[-0.779, -0.710]$	-0.158	-0.696
Width 32	-12.546, -12.659, -12.158	12, 12, 12	$[0.050, 0.087]$	0.115	0.286
Two experts	-12.532, -12.478, -12.228	11, 12, 12	$[0.014, 0.035]$	0.100	0.179
Curriculum	-12.676, -12.903, -12.209	11, 12, 12	$[0.153, 0.292]$	0.255	0.117

Initialization seeds are 9201, 9202 and 9203 in order. Intervals are unadjusted conditional 95% paired-case intervals, not intervals over training datasets or initialization seeds. Subgroup gains average all matching conditions; they do not imply improvement in every matching cell.

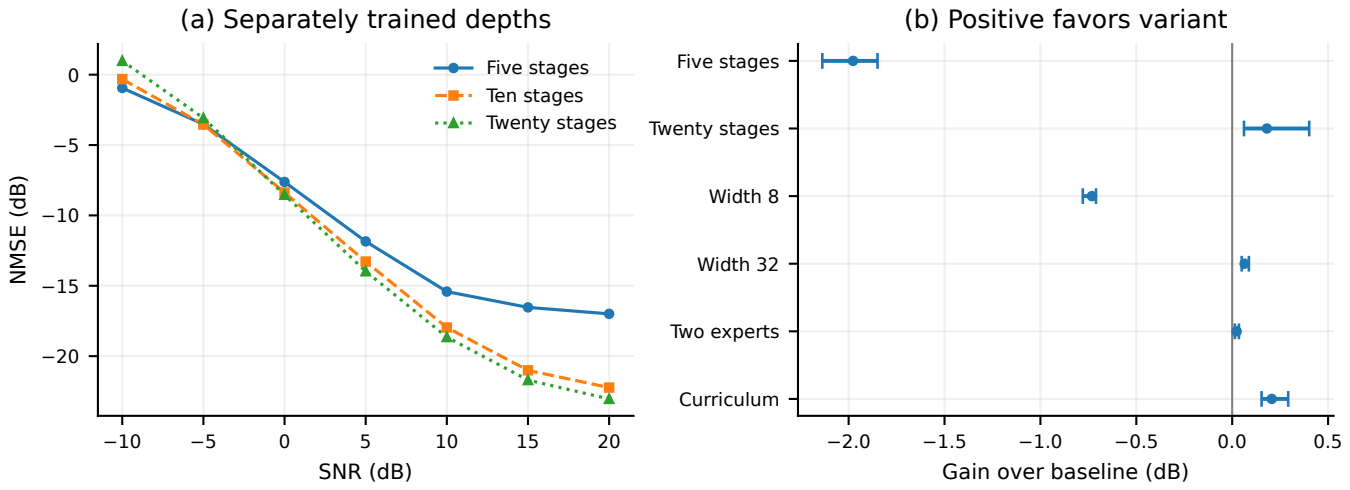


Fig. 11. New reduced-budget training study. Left: selected-model NMSE, averaging four reference levels, two path settings and three initializations at each SNR. Right: gain over ten stages/width 16 across all 56 conditions. Bars are unadjusted conditional 95% paired-case intervals; they exclude training-data and initialization uncertainty.

selected-model, 1,400 final-model and 112 classical files. Errors, severe-error rates and headline averages were recalculated from predictions and truth. Matching orders and masks were checked with the stated curriculum exception. One world per test condition was regenerated from its seed; selected and final models were replayed individually on a saved case. The largest replay difference was below 1.2×10^{-15}. Seventeen focused tests cover the new numerical/training/seed checks and OptShrink implementation. The release retains these audit tools and saved observations, not just summary scores. This checks the reported computation; it is not external replication or real-receiver validation.

XXIV. HISTORICAL: RELATED STRUCTURAL AND SPARSE METHODS

OptShrink estimates singular-value weights from the observed noise spectrum [1]. LRHD combines that principle with repeated Hankel projection [5]; neither our first one-pass diagnostic nor the new EM/OptShrink adaptation reproduces full LRHD. In both cases the starting problem is noisy matrix

denoising, not recovery from affine magnitudes. The noise on a Hankel matrix formed from an iterated channel estimate need not meet asymptotic independent-entry assumptions. Neither those guarantees nor universal optimality are asserted here.

Structured alternating least squares reconstructs low-rank matrices from linear measurements, including Hankel structure [6]. HSGD addresses partially observed Hankel matrices with sparse gross corruptions [7]. Their losses, observations and guarantees cannot be transferred unchanged to our nonlinear Rician model. They remain important alternative constructions, not irrelevant methods.

Noncoherent compressive channel estimation combines phase retrieval and sparse reconstruction using a designed sensing cascade [8]. GP-bandit beam training uses adaptive near-field beams, Rician-debiased pseudo-amplitudes and sparse phase retrieval [9]; it has no known additive reference. Partially coherent matching pursuit retains complex observations with a shared unknown phase within each packet [10]. That information is absent from our magnitude-only inputs. A fair benchmark must adapt these methods to the same observation and pilot budget rather than importing their published scores.

The multi-shot Rydberg study [11] concerns signal-presence detection with Rician shots. It motivates accounting for observation statistics and shot budgets, but supplies neither channel-estimation results nor a heavy-tailed noise model for this experiment.

XXV. Historical: What Is Still Not Tested

Depth. The new 5/10/20-stage comparison uses separately trained models, unlike the original prefix diagnostic. It has a fixed small training budget, not an exhaustive search or a convergence test. An optimal depth for larger datasets or other channel populations is not known.

Noise and hardware. The original model has independent complex Gaussian noise before magnitude extraction. Incorrect variance inputs do not test non-Gaussian noise. Finite shot count alone does not imply heavy-tailed noise, and no cited detection model establishes that assumption for this receiver. Optical readout, atomic dynamics and hardware data remain untested. A Rician loss is not automatically matched to an alternative optical or multi-shot measurement statistic.

Structure. Cubic shrinkage is one design choice, not a proved optimal denoiser. The new in-loop comparison covers one OptShrink-based reconstruction. Full LRHD, structured ALS and robust Hankel completion would require their own adapted measurement models and independently selected settings. Existing hard, global, adaptive and no-filtering controls do not exhaust these choices. An angle-domain sparse phase-retrieval method was also missing from this historical comparison, particularly for two-path channels; the current compact sparse/EM comparison addresses that gap. No superiority over the remaining untested methods, or over published URformer, is claimed.

Training. Two-path and RSR 5 dB cases already occur in the original training; RSR 0 dB does not. The new smaller study includes RSR 0 dB, and its curriculum has mixed results. No safety-critical protection guarantee follows from average gains. The original pointwise intervals condition on fixed models and shared training data; the fresh-data follow-up is limited and does not estimate broad training-population uncertainty.

Array geometry. A two-dimensional array may suggest multilevel or block-Hankel structure, but its lifting dimensions, normalization, computational cost and training transfer must be defined anew. No 2D array result is reported.

XXVI. Historical: Initialization and Numerical Record

For array element n, form $C_n \in \mathbb{C}^{P\times(K+1)}$ with rows $[s_p^T, b_{np}]$. Let u_n be the dominant eigenvector of $C_n^H \operatorname{diag}(z_n) C_n$ and set

$$c_n = \frac{\sum_p z_{np} |(C_n u_n)_p|}{\sum_p |(C_n u_n)_p|^2}. \tag{32}$$

The first K components of $c_n u_n \exp[-j \arg(u_{n,K+1})]$ initialize the channel row. In the original training recipe, only the 21 input pilots enter this calculation; withheld columns do not enter the initial estimate. The all-pilot alternative instead initializes from its 30-column noisy input view.

The original 43 focused tests cover likelihood and derivative checks, noise scaling, finite gradients, repeated singular values, pilot-split isolation, single/double precision agreement and deterministic recovery. The submission audit recalculated 1,120 main, 702 follow-up and 432 input-study result files. Three primary checkpoints replayed eight saved observations exactly; 24 input-study checkpoint replays were also checked. This is not a replay of every historical experiment or an independent hardware validation.

One finite-input SVD failure occurred in the original classical HS-GS test at $N = 64$. The same matrix was recomputed with SciPy's alternative SVD driver; relative reconstruction error was below 3×10^{-15}. The trial was retained with unchanged inputs and iteration budget. Original software versions were Python 3.12, NumPy 2.4.6, SciPy 1.17.1 and PyTorch 2.13.0. The saved local experimental record includes inputs, truth for evaluation, predictions, selected/final checkpoints, training histories, settings, source hashes and figure scripts for the reported studies. The separate Overleaf archive contains manuscript files, not experimental data. No GPU agreement or hardware measurement is claimed.

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